

Practical problems

— surface area and volume in context

The formulas are known; the whole skill is reading which one the situation wants.

LESSON 53–54

2 × 40 MIN

GEOMETRY 11, AMALIY MASALALAR

IGX 13.X IN CONTEXT

LESSON CLOCK

12'

20'

20'

20'

8'

Reading the question	12 min
Units	20 min
Density and rate	20 min
Which faces	20 min
Homework	8 min

LEARNING OBJECTIVES

- Choose between capacity, volume, curved area and total area from the wording.
- Convert between cm³, litres and m³ correctly in a single calculation.
- Use density and rate to turn a volume into a mass or a time.
- Report an answer to an accuracy the data supports.

TEXTBOOK

National	pp. 281–290
Cambridge	Core & Extended, pp. 379–388
Workbook	Exercise 13.6

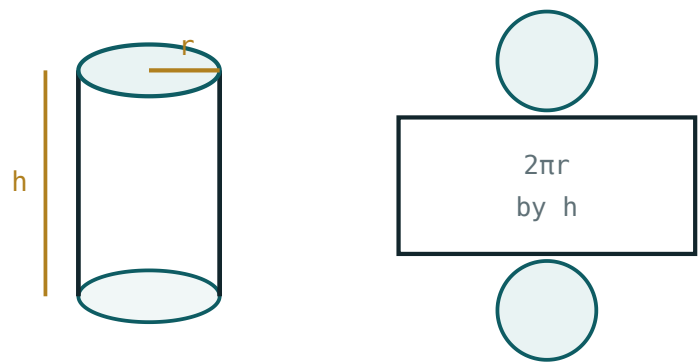
01

Explanation

Reading the question

Nothing in Quarter IV is new. The whole difficulty is that the question no longer says which formula it wants — it describes a situation, and you must decide.

THE QUESTION SAYS	IT WANTS	WATCH FOR
“how much will it hold”	capacity — volume from inner dimensions	litres, not cm ³
“how much metal”	volume of material — a difference	outer minus inner
“how much paint”	surface area	which faces are painted
“how much label”	curved surface only	no circles
“how heavy”	volume × density	units of density
“how long to fill”	volume ÷ rate	rate units



$S = 2\pi r^2 + 2\pi rh$ – two circles and one rectangle

A cylinder unrolled — the label is the rectangle, the lid and base the two circles.

THREE QUESTIONS BEFORE ANY ARITHMETIC

- Am I asked for a **volume** or an **area**?
- Is the object **open or closed**, **solid or hollow**?
- What **units** does the answer want?

Answer all three in writing and the calculation itself is Grade 11 Quarter II work.

Units

$1 \text{ litre} = 1000 \text{ cm}^3$ $1 \text{ m}^3 = 1000 \text{ litres} = 1\,000\,000 \text{ cm}^3$

GIVEN IN	ANSWER WANTED IN	MULTIPLY BY
cm^3	<i>litres</i>	$\div 1000$
m^3	<i>litres</i>	$\times 1000$
m^3	cm^3	$\times 10^6$
m^2	cm^2	$\times 10^4$

CONVERT THE LENGTHS FIRST, NOT THE ANSWER

A trough 2.5 m long with a cross-section in cm has two different units in one formula. Convert 2.5 m to 250 cm **before** multiplying; converting the volume afterwards needs 10^6 , and that is where the factor-of-a-thousand errors come from.

The squared and cubed conversion factors are the ones that catch people: $1\text{ m}^2 = 10\,000\text{ cm}^2$ and $1\text{ m}^3 = 1\,000\,000\text{ cm}^3$, not 100 and 100 .

Density and rate

$$mass = volume \times density \quad time = \frac{volume}{rate} \quad paint = \frac{area}{coverage}$$

QUANTITY	TYPICAL UNITS	EXAMPLE
density	g/cm^3	steel 7.8, water 1.0, aluminium 2.7
flow rate	$litres/min$	a tap 12 l/min
coverage	$m^2/litre$	paint 10 m^2/l

Worked chain. A steel pipe, outer radius 9 cm, inner 7 cm, length 2 m:

STEP	WORKING	VALUE
length in cm	$2\text{ m} = 200\text{ cm}$	200
cross-section	$\pi(81 - 49)$	$32\pi\text{ cm}^2$
volume	$32\pi \times 200$	$6400\pi \approx 20\,106\text{ cm}^3$
mass	$\times 7.8\text{ g/cm}^3$	$\approx 157\text{ kg}$

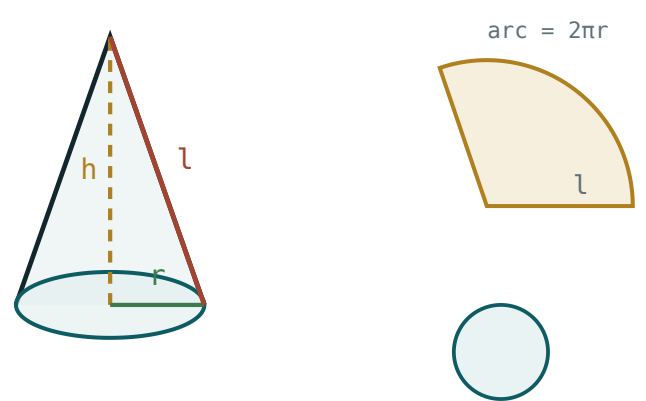
KEEP π UNTIL THE LAST LINE

6400π is exact; $20\,106$ is not. Rounding at the volume stage and again at the mass stage compounds the error. One rounding, at the end.

Which faces

Surface-area questions turn on which faces exist and which are wanted.

OBJECT	AREA WANTED	FORMULA
a label round a tin	curved only	$2\pi rh$
a closed tin	total	$2\pi r^2 + 2\pi rh$
an open cylindrical tank	base + curved	$\pi r^2 + 2\pi rh$
a silo painted above ground	curved + dome	$2\pi rh + 2\pi r^2$
a bowl, inside and out	both curved surfaces	$2\pi(R^2 + r^2) + rim$
a conical tent with a floor	curved + base	$\pi rl + \pi r^2$



$S = \pi r^2 + \pi rl$ — a circle and a sector

A cone unrolled — the sector is the curved surface, the circle the base.

THE COMMONEST SINGLE ERROR IN THE WHOLE QUARTER

Using $2\pi r^2 + 2\pi rh$ for an open tank. If the question says **open, tank, trough, bowl or tent without a floor**, at least one circle is missing. Read the noun, not just the numbers.

02 Worked examples

EXAMPLE 1 A cylindrical tank of radius 1.5 m and height 4 m is open at the top. Find its capacity in litres and the area to be painted inside and out (not the base).

$$1 \quad V = \pi(2.25)(4) = 9\pi \approx 28.27 \text{ m}^3.$$

$$2 \quad = 28\,270 \text{ litres.} \\ \times 1000.$$

$$3 \quad \text{Curved area } 2\pi(1.5)(4) = 12\pi \approx 37.7 \text{ m}^2, \text{ twice for inside and out.}$$

$$4 \quad \approx 75.4 \text{ m}^2.$$

ANSWER

$\approx 28\,300$ litres; $\approx 75.4 \text{ m}^2$

EXAMPLE 2 A steel pipe has outer radius 9 cm, inner 7 cm and length 2 m. Find its mass at 7.8 g/cm³.

$$1 \quad \text{Length } 200 \text{ cm.} \\ \text{Convert first.}$$

$$2 \quad V = \pi(81 - 49)(200) = 6400\pi$$

$$3 \quad \approx 20\,106 \text{ cm}^3.$$

$$4 \quad m \approx 156\,800 \text{ g} \approx 157 \text{ kg.}$$

ANSWER

$\approx 157 \text{ kg}$

EXAMPLE 3 A tap delivering 12 litres a minute fills a cuboid tank $1.2 \times 0.8 \times 0.9$ m. How long does it take?

① $V = 0.864 \text{ m}^3$.

② $= 864$ litres.

③ $t = \frac{864}{12} = 72$ minutes.

④ That is 1 h 12 min.

ANSWER

72 minutes

EXAMPLE 4 A conical tent has base radius 3 m and slant height 5 m, with a groundsheet. Find the canvas needed and the air space inside.

① Curved $\pi r l = 15\pi \approx 47.1 \text{ m}^2$.

② Groundsheet $\pi r^2 = 9\pi \approx 28.3 \text{ m}^2$.

③ Total $24\pi \approx 75.4 \text{ m}^2$.

④ Height $\sqrt{25 - 9} = 4$; $V = \frac{1}{3}\pi(9)(4) = 12\pi \approx 37.7 \text{ m}^3$.

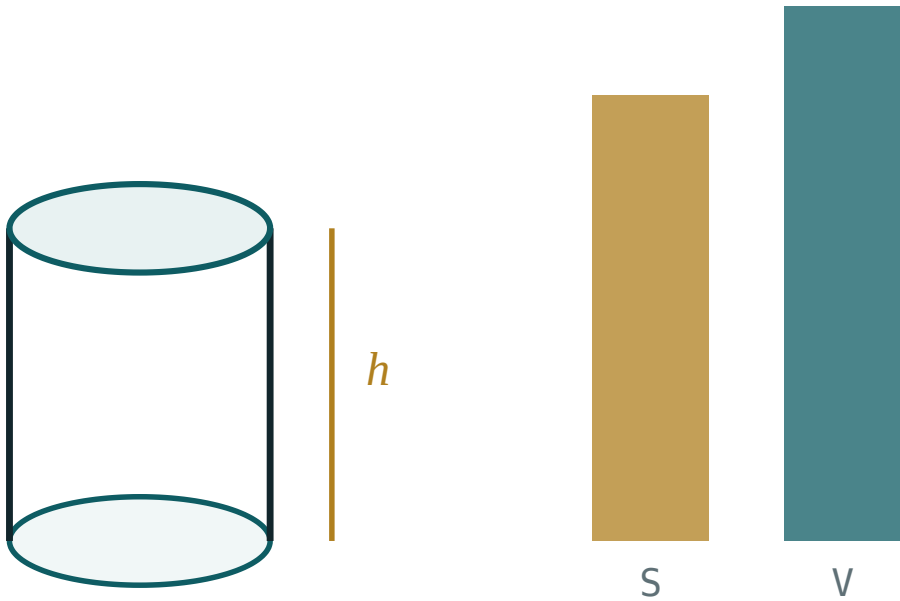
ANSWER

$\approx 75.4 \text{ m}^2$ of material; $\approx 37.7 \text{ m}^3$

03 Interactive model

Bring a tin, a bottle and a funnel; the class decides for each which formula the label, the contents and the metal need.

A cylinder, and what it holds



base area B **50.3**

perimeter P **25.1**

total S **251.3**

volume V **301.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Capacity	Sig'im	Вместимость
Volume	Hajm	Объём
Density	Zichlik	Плотность
Flow rate	Oqim tezligi	Скорость потока
Wetted surface	Ho'llanadigan sirt	Смачиваемая поверхность
Coverage	Sarf	Расход
Open solid	Ochiq jism	Открытое тело
Litre	Litr	Литр
Unit conversion	Birliklarni almashtirish	Перевод единиц
Degree of accuracy	Aniqlik darajasi	Степень точности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“How much will it hold” asks for:

surface area

capacity

mass

the curved area

1 m³ in litres:

100

1000

10 000

1 000 000

1 m² in cm²:

100

1000

10 000

1 000 000

An open cylindrical tank has area:

$$2\pi r^2 + 2\pi rh$$

$$\pi r^2 + 2\pi rh$$

$$2\pi rh$$

$$\pi r^2$$

A label on a tin needs:

$$2\pi rh$$

$$2\pi r^2 + 2\pi rh$$

$$\pi r^2$$

$$\pi rl$$

Mass equals:

$$\text{volume} \div \text{density}$$

$$\text{volume} \times \text{density}$$

$$\text{area} \times \text{density}$$

$$\text{density} \div \text{volume}$$

Rounding should happen:

at each step

once, at the end

never

twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Capacity of a cylinder $r = 2$, $h = 5$ cm, in cm^3

ANSWER $20\pi \approx 62.8$

2. Same, in litres

ANSWER ≈ 0.0628

3. 1 m^3 in litres

ANSWER 1000

4. 2.5 m in cm

ANSWER 250

5. Label area for $r = 3$, $h = 10$

ANSWER $60\pi \approx 188$

6. Closed tin total area, $r = 3$, $h = 10$

ANSWER $78\pi \approx 245$

7. Mass of 1000 cm^3 of steel at 7.8 g/cm^3

ANSWER 7.8 kg

Medium · the standard question

7 PROBLEMS

1. Open tank $r = 1.5$, $h = 4$ m: capacity in litres

ANSWER $\approx 28\,300$

2. Same tank: area painted inside and out, base excluded

ANSWER $\approx 75.4\text{ m}^2$

3. Pipe $R = 9$, $r = 7$ cm, $L = 2$ m: volume of metal

ANSWER $6400\pi \approx 20\,100\text{ cm}^3$

4. Same pipe: mass at 7.8 g/cm^3

ANSWER $\approx 157\text{ kg}$

5. Tank $1.2 \times 0.8 \times 0.9$ m at 12 l/min : filling time

ANSWER 72 min

6. Conical tent $r = 3$, $l = 5$ with floor: material

ANSWER $24\pi \approx 75.4\text{ m}^2$

7. Same tent: air space

ANSWER $12\pi \approx 37.7\text{ m}^3$

Hard · stretch and reason

7 PROBLEMS

1. A silo, cylinder $r = 4$, $h = 10$ m with a hemispherical top: volume and painted area above ground

ANSWER $\frac{608\pi}{3} \approx 637 \text{ m}^3$; $112\pi \approx 352 \text{ m}^2$

2. Paint at $10 \text{ m}^2/\text{litre}$ for that silo: litres needed

ANSWER $\approx 36 \text{ litres}$

3. A hemispherical bowl of inner radius 12 cm : capacity in litres

ANSWER $\approx 3.62 \text{ litres}$

4. A trough with trapezium section $40/60 \text{ cm}$, depth 30 cm , length 3 m : capacity in litres

ANSWER 450 litres

5. A gutter is half a cylinder of radius 8 cm and length 6 m : material area

ANSWER $4800\pi \text{ cm}^2 \approx 1.51 \text{ m}^2$

6. A tank of 2000 litres is a cylinder with $h = 2r$: find r

ANSWER $r = \sqrt[3]{\frac{1}{\pi}} \approx 0.683 \text{ m}$

7. A ball of radius 20 cm is painted at 2 so'm per cm^2 : the cost

ANSWER $3200\pi \approx 10\,050 \text{ so'm}$

07 Homework

Homework — 5 tasks

Answer the three reading questions in writing before every calculation.

- A cylindrical water tank is open at the top, radius 1.2 m , height 3 m . Find its capacity in litres and the area of metal used.
- A copper pipe has outer radius 5 cm , inner 4 cm and length 3 m . Find the mass of copper at 8.9 g/cm^3 .

3. A tap delivers 15 litres a minute into a tank $1.5 \times 1.0 \times 1.2$ m. Find the time to fill it.
4. A conical tent of base radius 2.5 m and height 6 m has no groundsheet. Find the canvas needed and the volume inside.
5. For each of a tin label, an open trough and a solid steel ball, say which formula is needed and why, in one sentence each.

Optimisation of a container [Cambridge insert]

One constraint, one variable, one derivative — and the cheapest tin in the shop.

LESSON 55–56

2 × 40 MIN

GEOMETRY 11, AMALIY MASALALAR

P1 9.3

LESSON CLOCK

11'

23'

19'

19'

8'

The four steps	11 min
The closed cylinder	23 min
Open containers	19 min
Cost, not area	19 min
Homework	8 min

LEARNING OBJECTIVES

- Form an expression for the quantity to be optimised.
- Use the constraint to reduce it to one variable.
- Differentiate, solve $f' = 0$ and verify the nature of the stationary point.
- Interpret the answer, including any practical restriction.

TEXTBOOK

National	pp. 291–300
Cambridge	Pure Mathematics 1, pp. 196–204
Workbook	Exercise 9C

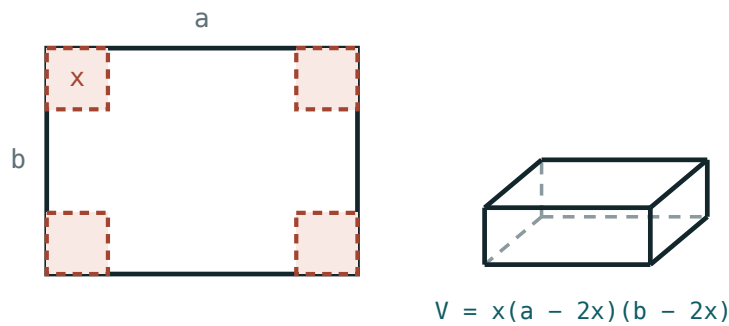
01

Explanation

The four steps

EVERY OPTIMISATION QUESTION, IN FOUR STEPS

- Write the quantity to be optimised (the **objective**) in terms of the dimensions.
- Write the **constraint**, and use it to eliminate one variable.
- Differentiate, set $= 0$, solve.
- Verify it is a minimum (or maximum) and answer the question asked.



cut a square x from each corner

The classic box — one constraint, one variable, one derivative.

STEP 4 IS A REAL MARK

Finding r is not the answer if the question asked for the **area**, or for h , or for the ratio $\frac{h}{r}$. And a stationary point must be shown to be a minimum, by $f'' > 0$ or a sign change of f' .

The closed cylinder

The standard question: a closed cylindrical tin of fixed volume V — what shape uses least metal?

STEP	WORKING
objective	$S = 2\pi r^2 + 2\pi rh$
constraint	$V = \pi r^2 h \Rightarrow h = \frac{V}{\pi r^2}$
eliminate	$S = 2\pi r^2 + \frac{2V}{r}$
differentiate	$\frac{dS}{dr} = 4\pi r - \frac{2V}{r^2}$
solve	$4\pi r^3 = 2V \Rightarrow r = \sqrt[3]{\frac{V}{2\pi}}$
verify	$\frac{d^2S}{dr^2} = 4\pi + \frac{4V}{r^3} > 0 — a\ minimum$

THE ANSWER IN ONE SENTENCE

Substituting back gives $h = 2r$: **the least-metal closed tin is as tall as it is wide**. Its height equals its diameter, and it fits exactly inside a cube. The result does not depend on V at all.

Example. $V = 2000\text{ cm}^3$ gives $r = \sqrt[3]{\frac{2000}{2\pi}} \approx 6.83\text{ cm}$ and $h \approx 13.66\text{ cm}$, with $S \approx 879\text{ cm}^2$. A tin of the same volume with $r = 4$ needs $S \approx 1101\text{ cm}^2$ — a quarter more metal for the same contents.

Open containers

Remove a face and the objective changes, so the answer does too.

CONTAINER	OBJECTIVE	OPTIMAL SHAPE
closed cylinder	$2\pi r^2 + 2\pi rh$	$h = 2r$
open cylinder (no lid)	$\pi r^2 + 2\pi rh$	$h = r$
closed cuboid, square base	$2x^2 + 4xh$	$h = x$
open cuboid, square base	$x^2 + 4xh$	$h = \frac{x}{2}$

Every one is the same four steps. Note the pattern: removing the lid makes the optimal container **shallower**, because height is now relatively more expensive than base.

A CUBE IS OPTIMAL ONLY WHEN THE BOX IS CLOSED

The closed cuboid of least area for a given volume is a cube. The open one is **not** — it is half as tall as it is wide. Assuming a cube without checking is a common and costly shortcut.

Cost, not area

When materials differ in price, minimise the **cost**, not the area — the four steps are unchanged, but the objective carries the prices.

Example. An open tank with a square base of side x and height h holds 32 m^3 . The base costs 10 units per m^2 , the sides 5 :

$$C = 10x^2 + 5(4xh) = 10x^2 + 20xh, h = \frac{32}{x^2}$$

$$C = 10x^2 + \frac{640}{x}, \quad \frac{dC}{dx} = 20x - \frac{640}{x^2} = 0$$

So $x^3 = 32$, $x \approx 3.17$ m, $h \approx 3.17$ m — and here $h = x$, not $\frac{x}{2}$, because the base is twice as expensive and the design compensates by making it smaller.

CHECK THE FEASIBLE RANGE

A dimension must be positive, and often has an upper bound from the physical situation (a sheet of given size, a shelf of given height). If the stationary point lies outside that range, the optimum is at an **endpoint** — and the derivative never finds it. State the range before differentiating.

02 Worked examples

EXAMPLE 1 A closed cylindrical tin holds 2000 cm^3 . Find the radius and height that use least metal.

$$1 \quad S = 2\pi r^2 + \frac{4000}{r}$$

After eliminating h .

$$2 \quad \frac{dS}{dr} = 4\pi r - \frac{4000}{r^2} = 0$$

$$3 \quad r^3 = \frac{1000}{\pi} \Rightarrow r \approx 6.83$$

$$4 \quad h = \frac{2000}{\pi(46.6)} \approx 13.66 = 2r$$

$S'' > 0$, so a minimum.

ANSWER

$r \approx 6.83 \text{ cm}$, $h \approx 13.66 \text{ cm}$

EXAMPLE 2 An open cylindrical tank holds $54\pi \text{ cm}^3$. Find the radius that uses least metal.

$$(1) \quad S = \pi r^2 + 2\pi rh, \quad h = \frac{54}{r^2}$$

$$(2) \quad S = \pi r^2 + \frac{108\pi}{r}$$

$$(3) \quad \frac{dS}{dr} = 2\pi r - \frac{108\pi}{r^2} = 0 \Rightarrow r^3 = 54$$

$$(4) \quad r \approx 3.78, \quad h \approx 3.78 = r.$$

ANSWER

$$r = \sqrt[3]{54} \approx 3.78 \text{ cm, with } h = r$$

EXAMPLE 3 An open tank with a square base holds 32 m^3 . The base costs 10 and the sides 5 units per m^2 . Find the cheapest dimensions.

$$(1) \quad C = 10x^2 + 20xh, \quad h = \frac{32}{x^2}$$

$$(2) \quad C = 10x^2 + \frac{640}{x}$$

$$(3) \quad \frac{dC}{dx} = 20x - \frac{640}{x^2} = 0 \Rightarrow x^3 = 32$$

$$(4) \quad x \approx 3.17 \text{ m, } h \approx 3.17 \text{ m; } C \approx 302 \text{ units.}$$

ANSWER

$$x = h \approx 3.17 \text{ m}$$

EXAMPLE A rectangular sheet 40×40 cm has equal squares of side x cut from the corners; the flaps fold up.
 4 Find x for the greatest volume.

1 $V = x(40 - 2x)^2$, with $0 < x < 20$.

State the range.

2 $\frac{dV}{dx} = (40 - 2x)^2 - 4x(40 - 2x)$

3 $(40 - 2x)(40 - 6x) = 0 \Rightarrow x = 20$ (rejected) or $x = \frac{20}{3}$

4 $x \approx 6.67$ cm, $V \approx 4741$ cm³.

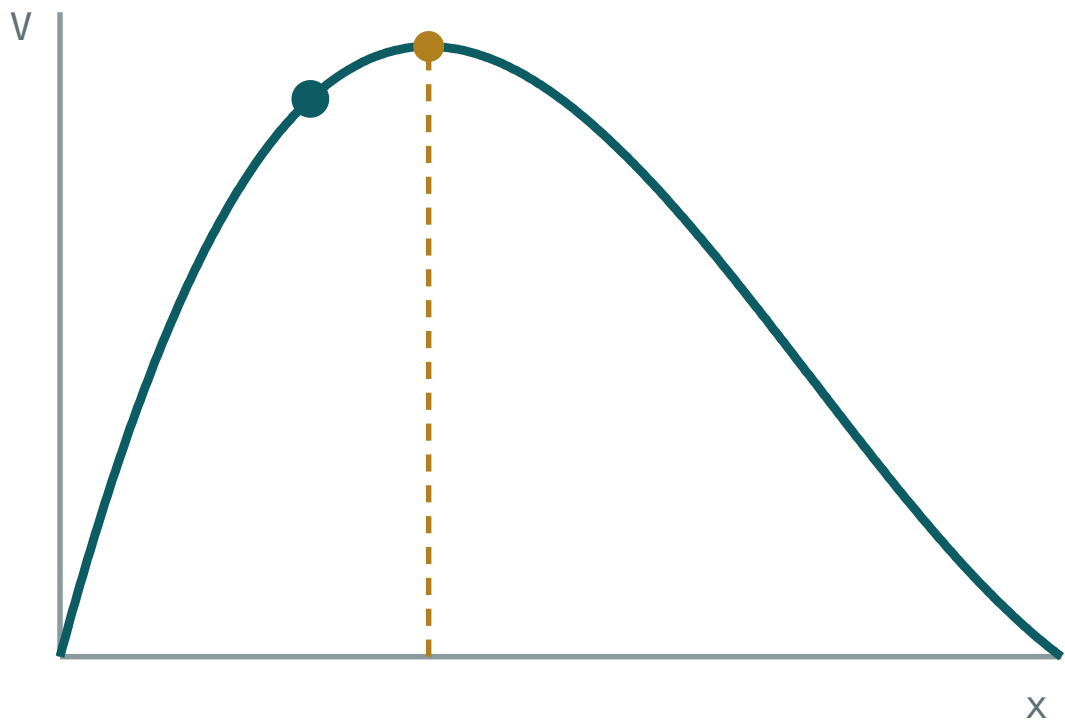
ANSWER

$$x = \frac{20}{3} \approx 6.67 \text{ cm}$$

03 Interactive model

Bring three tins of the same volume but different shapes; the class predicts which uses least metal, then measures.

The cheapest container



x 2

$V(x)$ 384

greatest V 420.1

at $x = 2.94$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Optimisation	Optimallashtirish	Оптимизация
Constraint	Cheklov	Ограничение
Objective function	Maqsad funksiyasi	Целевая функция
Stationary point	Statsionar nuqta	Стационарная точка
Minimum	Minimum	Минимум
Maximum	Maksimum	Максимум
Second derivative	Ikkinchi hosila	Вторая производная
Feasible range	Mumkin bo'lgan oraliq	Допустимый интервал
Cost function	Narx funksiyasi	Функция стоимости
Aspect ratio	Nisbat	Отношение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The constraint is used to:

check the answer

eliminate a variable

differentiate

nothing

A minimum is confirmed by:

$$f' = 0$$

$$f'' > 0$$

$$f'' < 0$$

$$f = 0$$

A closed cylinder of least area has:

$$h = r$$

$$h = 2r$$

$$h = \frac{r}{2}$$

$$h = 4r$$

An open cylinder of least area has:

$$h = r$$

$$h = 2r$$

$$h = \frac{r}{2}$$

$$h = 3r$$

A closed cuboid of least area is:

a cube

twice as tall as wide

half as tall as wide

any shape

An open cuboid of least area has:

$$h = x$$

$$h = \frac{x}{2}$$

$$h = 2x$$

$$h = 4x$$

If the stationary point is outside the feasible range:

there is no answer

the optimum is at an endpoint

differentiate again

ignore the range

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Objective for a closed cylinder

ANSWER $2\pi r^2 + 2\pi rh$

2. Objective for an open cylinder

ANSWER $\pi r^2 + 2\pi rh$

3. Constraint for a cylinder of volume V

ANSWER $h = \frac{V}{\pi r^2}$

4. Optimal shape of a closed cylinder

ANSWER $h = 2r$

5. Optimal shape of an open cylinder

ANSWER $h = r$

6. Optimal closed cuboid with square base

ANSWER $h = x$ — a cube

7. Test for a minimum

ANSWER $f'' > 0$

Medium · the standard question

7 PROBLEMS

1. Closed tin of 2000 cm^3 : the radius

ANSWER $\approx 6.83 \text{ cm}$

2. Same tin: the height

ANSWER $\approx 13.66 \text{ cm}$

3. Same tin: the least area

ANSWER $\approx 879 \text{ cm}^2$

4. Open tank of $54\pi \text{ cm}^3$: the radius

ANSWER $\sqrt[3]{54} \approx 3.78$

5. Open cuboid, square base, volume 108 : the base edge

ANSWER $x = 6$

6. Box from a 40×40 sheet: the cut x

ANSWER $\frac{20}{3} \approx 6.67$

7. Same box: the greatest volume

ANSWER $\approx 4741 \text{ cm}^3$

Hard · stretch and reason

7 PROBLEMS

1. Open tank, 32 m^3 , base 10 and sides 5 per m^2 : the base edge

ANSWER $\approx 3.17 \text{ m}$

2. Same: the least cost

ANSWER $\approx 302 \text{ units}$

3. A cylinder inscribed in a sphere of radius R : the greatest volume

ANSWER $\frac{4\pi R^3}{3\sqrt{3}}$

4. A cone of slant l : the greatest volume

ANSWER $h = \frac{l}{\sqrt{3}}$

5. A closed cylinder where the ends cost twice the side: the optimal $\frac{h}{r}$

ANSWER 4

6. A box from a 30×50 sheet: the cut x for greatest volume

ANSWER $\approx 6.07 \text{ cm}$

7. A cylinder of volume V inscribed in a cone of radius R and height H : the greatest V

ANSWER $\frac{4\pi R^2 H}{27}$

07 Homework

Homework — 5 tasks

Write the four steps as headings and fill them in; the structure carries the method marks.

1. A closed cylindrical tin holds 1000 cm^3 . Find the radius and height that use least metal, and verify the minimum.
2. An open cylindrical tank holds $16\pi \text{ m}^3$. Find the radius that uses least material.
3. An open box with a square base holds 500 cm^3 . Find the dimensions of least surface area.

4. Squares of side x are cut from the corners of a 24×36 cm sheet and the flaps folded up. Find x for the greatest volume, stating the feasible range.
5. An open tank of 18 m^3 has a square base costing 8 and sides costing 4 units per m^2 . Find the cheapest dimensions.

Test practice — the state final attestation format

Thirty questions, forty minutes, no working shown — a different skill from the one practised all year.

LESSON 57–58

2 × 40 MIN

GEOMETRY 11, DTM FORMATI

EXAMINATION TECHNIQUE

LESSON CLOCK

8'

32'

16'

16'

8'

The format	8 min
A practice block of 20	32 min
Marking and analysis	16 min
The four fast techniques	16 min
Homework	8 min

LEARNING OBJECTIVES

- Work a multiple-choice geometry question in under 80 seconds.
- Use elimination, estimation and special cases when a full solution is too slow.
- Recognise the distractors that mark schemes deliberately include.
- Manage the whole paper: order, timing and the last five minutes.

TEXTBOOK

National	pp. 301–310
Cambridge	Core & Extended, pp. 389–392
Workbook	Test paper G1

01

Explanation

The format

FEATURE	WHAT IT MEANS FOR YOU
30 questions, 40 minutes	80 seconds each, on average
four options, one correct	elimination is a legitimate method
no working marked	a correct guess scores the same as a proof
no negative marking	never leave a question blank
calculator usually not allowed	answers are designed to be exact

THE ORDER TO WORK IN

1. **First pass** (20 min): every question you can do in under a minute. Mark the rest.

2. Second pass (15 min): the marked ones, hardest last.

3. Last 5 minutes: transfer everything to the answer sheet, and fill every blank with your best elimination.

A question you cannot do in 80 seconds costs you two you could have done. Leaving it is not giving up; it is arithmetic.

The four fast techniques

1. Elimination by size. A volume must be positive, a probability at most 1, an angle in a triangle under 180° , a surface area larger than any one face. Two of four options usually fail one of these tests.

2. Estimation. $\pi \approx 3$, $\sqrt{2} \approx 1.4$, $\sqrt{3} \approx 1.7$. A cone of radius 5 and height 12 has volume about $\frac{1}{3}(3)(25)(12) = 300$ — enough to pick between 100π and 300π .

3. Special cases. If the answer must hold for every triangle, test the equilateral one. If for every n , test $n = 1$ and $n = 2$. An option that fails a special case is out.

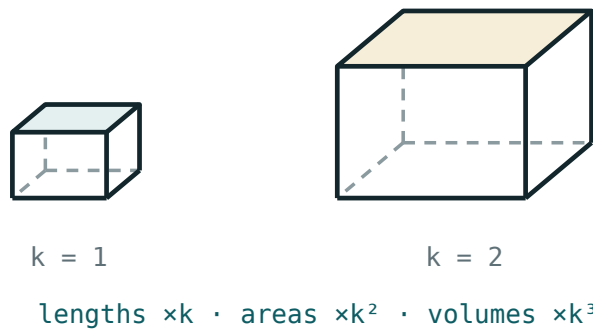
4. Work backwards from the options. Substituting four given answers is often faster than solving. If the options for a radius are 2, 3, 4, 6, computing V for each takes less time than solving a cubic.

QUESTION	FAST ROUTE	TIME
volume of a cone, $r = 5$, $h = 12$	estimate ≈ 314	10 s
surface area of a sphere of radius 3	$4\pi(9) = 36\pi$	5 s
a ratio of similar solids 8 : 27	cube roots 2 : 3	10 s
which solid has $V = \frac{1}{3}Bh$	recall	3 s

The distractors

Wrong options are not random. Each one is the answer to a specific mistake, which means you can recognise your own error in the list.

THE DISTRACTOR IS	THE MISTAKE IT CATCHES
three times the right answer	the missing $\frac{1}{3}$ in a cone or pyramid
half or double	radius used for diameter
the curved area when total was asked	reading
k instead of k^2 or k^3	similar-solids ratios
the slant instead of the height	a cone volume
an answer in cm^3 when litres were asked	units



k, k^2, k^3 — three of the four options in every similarity question.

IF YOUR ANSWER IS EXACTLY ONE OF THE OPTIONS, CHECK ANYWAY

Finding your answer in the list is **not** confirmation — the distractors were built from the likely errors, so a wrong answer is more likely to appear than a random number would be. Spend the last five seconds on the reading check: total or curved? radius or diameter? cm^3 or litres?

A practice block

Twenty questions, twenty-five minutes. Work them in the order given, marking anything that takes longer than a minute.

#	QUESTION	ANSWER
1	Volume of a cube of edge 5	125
2	Space diagonal of that cube	$5\sqrt{3}$
3	Surface area of a sphere of radius 3	36π
4	Volume of a cone, $r = 3$, $h = 4$	12π
5	Slant height of that cone	5
6	Curved area of that cone	15π
7	Volume of a sphere of radius 6	288π
8	Volume of a pyramid, $B = 36$, $h = 5$	60
9	Two similar solids, volumes 8 : 27: length ratio	2 : 3
10	Same pair: area ratio	4 : 9
11	Volume of a cylinder, $r = 2$, $h = 7$	28π
12	Total area of that cylinder	36π
13	Angle of a cube's space diagonal with the base	$\approx 35.3^\circ$
14	Frustum volume, $R = 4$, $r = 2$, $h = 3$	28π
15	Sector angle of the net of a cone, $r = 3$, $l = 9$	120°
16	Capacity in litres of 2000 cm^3	2
17	Curved area of a hemisphere of radius 5	50π
18	Volume of a prism, $B = 15$, $h = 8$	120
19	A cap of height 2 on a sphere of radius 5: its curved area	20π
20	Closed cylinder of least area for a given V: $\frac{h}{r}$	2

MARK IT, THEN COUNT THE TYPES

Do not count the score. Count how many were lost to **reading**, how many to **a formula**, and how many to **arithmetic**. Those three numbers say what to practise; the score says nothing.

02 Worked examples

EXAMPLE 1 A cone has $r = 5$ and $h = 12$. Which is its volume: 100π , 300π , 60π or 900π ?

$$(1) \quad V = \frac{1}{3}\pi(25)(12)$$

$$(2) \quad = 100\pi$$

$$(3) \quad \text{Note the distractor } 300\pi \text{ — the missing } \frac{1}{3}.$$

ANSWER

 100π

EXAMPLE 2 Two similar cones have volumes in the ratio $8 : 27$. What is the ratio of their curved areas?

$$(1) \quad k^3 = \frac{27}{8} \Rightarrow k = \frac{3}{2}$$

Cube root.

$$(2) \quad k^2 = \frac{9}{4}$$

$$(3) \quad 4 : 9$$

The distractors would be $2 : 3$ and $8 : 27$.

ANSWER

 $4 : 9$

EXAMPLE 3 A sphere has surface area 100π . Its volume is: $\frac{500\pi}{3}$, 500π , $\frac{250\pi}{3}$ or 100π ?

① $4\pi R^2 = 100\pi \Rightarrow R = 5$

② $V = \frac{4}{3}\pi(125) = \frac{500\pi}{3}$

③ 500π is the missing $\frac{4}{3}$ distractor.

ANSWER

$$\frac{500\pi}{3}$$

EXAMPLE 4 Without computing exactly, decide whether a cylinder with $r = 7$ and $h = 10$ holds more or less than 1.5 m^3 .

① Units: r and h in cm — read again.

② If in cm: $V \approx 3(49)(10) \approx 1470 \text{ cm}^3$.

③ That is 1.47 litres, not m^3 .

The unit trap, caught by estimation.

ANSWER

About 1.5 litres — the units matter more than the arithmetic

03 Interactive model

Time the class strictly on the twenty questions; the discussion afterwards is about pace, not about geometry.

Twenty questions against the clock

Volume of a cube of edge 5:

Space diagonal of that cube:

Surface area of a sphere, $R = 3$:

Volume of a cone, $r = 3$, $h = 4$:

Curved area of that cone:

Volume of a sphere, $R = 6$:

Similar solids, volumes $8 : 27$: length ratio

Same pair: area ratio

$2 : 3$

$4 : 9$

$8 : 27$

$16 : 81$

Total area of a cylinder, $r = 2$, $h = 7$:

28π

36π

32π

44π

2000 cm^3 in litres:

0.2

2

20

200

Sector angle of a cone's net, $r = 3$, $l = 9$:

90°

120°

180°

60°

Least-area closed cylinder has h r :

1

2

3

$\frac{1}{2}$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Attestation (final test)	Yakuniy attestatsiya	Итоговая аттестация
Multiple choice	Test savoli	Тестовый вопрос
Distractor	Chalg‘ituvchi javob	Отвлекающий ответ
Elimination	Chiqarib tashlash	Исключение
Estimation	Baholash	Оценка
Special case	Xususiy hol	Частный случай
Time per question	Savolga vaqt	Время на вопрос
Answer sheet	Javoblar varaqasi	Бланк ответов

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

With no negative marking you should:

leave hard questions blank

answer every question

guess only at the end

skip the last five

A question taking over 80 seconds should be:

finished anyway

marked and left for the second pass

guessed at once

skipped for ever

A distractor three times the right answer catches:

a unit error

the missing $\frac{1}{3}$

a reading error

nothing

Finding your answer among the options means:

it is right

nothing on its own

check nothing

move on

After marking a practice block, count:

the score

reading, formula and arithmetic errors

the time

the blanks

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume of a cube of edge 5

ANSWER 125

2. Space diagonal of that cube

ANSWER $5\sqrt{3}$

3. Surface area of a sphere, $R = 3$

ANSWER 36π

4. Volume of a cone, $r = 3$, $h = 4$

ANSWER 12π

5. Slant height of that cone

ANSWER 5

6. Volume of a prism, $B = 15$, $h = 8$

ANSWER 120

7. 2000 cm^3 in litres

ANSWER 2

Medium · the standard question

7 PROBLEMS

1. Curved area of a cone, $r = 3$, $h = 4$

ANSWER 15π

2. Volume of a sphere, $R = 6$

ANSWER 288π

3. Total area of a cylinder, $r = 2$, $h = 7$

ANSWER 36π

4. Frustum volume, $R = 4$, $r = 2$, $h = 3$

ANSWER 28π

5. Sector angle of a cone net, $r = 3$, $l = 9$

ANSWER 120°

6. Curved area of a hemisphere, $R = 5$

ANSWER 50π

7. Cap of height 2 on a sphere of radius 5: curved area

ANSWER 20π

Hard · stretch and reason

7 PROBLEMS

1. Angle of a cube's space diagonal with the base

ANSWER $\approx 35.3^\circ$

2. Sphere of area 100π : its volume

ANSWER $\frac{500\pi}{3}$

3. Similar solids of volumes $8 : 27$: the area ratio

ANSWER $4 : 9$

4. Least-area closed cylinder: $\frac{h}{r}$

ANSWER 2

5. A cone of radius 5 and height 12 : total surface area

ANSWER 90π

6. A hemisphere and a cone on the same base of radius r have equal volumes: find h

ANSWER $h = 2r$

7. A cube and a sphere have equal surface areas: the ratio of their volumes (cube : sphere)

ANSWER $1 : \sqrt{\frac{6}{\pi}} \approx 1 : 1.38$

07 Homework

Homework — 4 tasks

Time yourself honestly; the timing is the point of the exercise, not the answers.

1. Work the twenty-question block again under timed conditions and record how many took more than 80 seconds.
2. For each question you got wrong, say whether the error was in reading, in the formula, or in the arithmetic.
3. Write out the six distractor types with one example of each from the block.

4. Write your plan for the 40 minutes of the real paper in three sentences: first pass, second pass, last five minutes.

Control work 4, and work on the mistakes

Every solid of the year in one paper, then a named diagnosis of each slip.

LESSON 59–60

2 × 40 MIN

GEOMETRY 11, NAZORAT ISHI 4

IGX 13 REVIEW

LESSON CLOCK

3

42'

11'

19'

5'

Instructions	3 min
The paper	42 min
Answers	11 min
Diagnosis and rewrite	19 min
The map	5 min

LEARNING OBJECTIVES

- Apply the surface-area and volume formulas of the whole year in one assessment.
- Handle compound solids, bounds and context correctly.
- Classify each lost mark and rewrite the solution in full.
- Draw the year’s mensuration as a single map.

TEXTBOOK

National	pp. 311–314
Cambridge	Core & Extended, pp. 393–396
Workbook	Control paper G4

01

Explanation

The paper — 40 marks, 45 minutes

Q	TASK	MARKS	FROM
1	A regular hexagonal prism, edge 4, height 9: total area and volume	6	Q II
2	A square pyramid, base edge 12, height 8: apothem, lateral area, volume	7	Q III
3	A cone of radius 6 and height 8: slant, curved area, volume, sector angle of the net	7	Q III
4	A ball of volume 972π : radius and surface area; then a cap of height 3	6	Q III
5	A silo — cylinder $r = 3$, $h = 8$ m plus a hemisphere: volume and painted area, base excluded	7	Q III–IV
6	A closed cylindrical tin of 500 cm^3 : the radius of least metal, verified	7	Q IV

WHERE THE MARKS ACTUALLY GO

Q2 carries one mark for using the apothem in the area and the height in the volume; Q3 one for the sector angle; Q5 one for excluding the covered circles; Q6 one for verifying the minimum. Four of the forty marks are for one decision each.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
slant for height	$V = \frac{1}{3}Bl$	volume uses the perpendicular
height for slant	$S = \frac{1}{2}Ph$	area uses the apothem
missing $\frac{1}{3}$	$V = Bh$	pyramids and cones are a third
sphere formulas swapped	$S = \frac{4}{3}\pi R^3$	S has R^2 , V has R^3
joined faces counted	adding two whole surface areas	subtract twice the shared face
ratios not cubed	$V_1 : V_2 = k$	k, k^2, k^3
minimum not verified	$f' = 0$	add $f'' > 0$

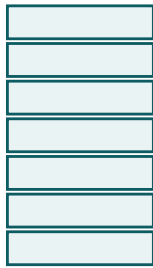
Write the slip's name in the margin of every question that lost a mark, then rewrite the whole solution — not the wrong line.

The year as one map

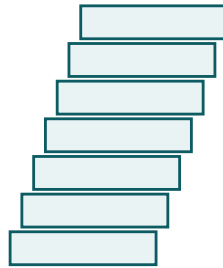
Eight boxes, links written as sentences:

- **coordinates in space** → **vectors** — “a point becomes three numbers”
- **vectors** → **the scalar product** — “angles without a picture”
- **prism** → $V = Bh$ — “uniform cross-section”
- **cylinder** → **the same, with $B = \pi r^2$**
- **pyramid and cone** → **a third of it** — “the same base, the same height”
- **sphere** → **2 : 3 with its cylinder** — “Archimedes, for both”
- **compound** → **add or subtract, and drop the joined faces**
- **scale k** → k, k^2, k^3 — “length, area, volume”

same base, same height, same volume



right prism



oblique prism

every horizontal slice has the same area

Equal slices, equal volumes — the principle under half the boxes above.

NINE FORMULAS, ONE PAGE

$$V_{prism} = Bh \quad V_{cyl} = \pi r^2 h \quad V_{pyr} = \frac{1}{3} Bh \quad V_{cone} = \frac{1}{3} \pi r^2 h \quad V_{sph} = \frac{4}{3} \pi R^3$$

$$S_{cyl} = 2\pi r(r + h) \quad S_{cone} = \pi r(r + l) \quad S_{sph} = 4\pi R^2 \quad V_{frus} = \frac{\pi h}{3} (R^2 + Rr + r^2)$$

If these nine sit on one page in your own handwriting, the annual review needs nothing new.

Looking forward

Lessons 61–68 are pure revision: coordinates and vectors, parallelism and perpendicularity, all the surfaces and volumes, then the entrance paper and the annual review. Nothing is added; the work is to make what exists fast and certain.

ONE HABIT TO CARRY FORWARD

The whole year rewarded writing B , h and the shape's name before any arithmetic. Keep that habit into the entrance paper, where the question rarely says which solid it is describing and the first line of the answer has to.

02 Worked examples

EXAMPLE 1 Model answer, Q2: a square pyramid, base edge 12, height 8.

① Base apothem 6; $m = \sqrt{64 + 36} = 10$.

The first right triangle.

② $S_{lat} = \frac{1}{2}(48)(10) = 240$

Apothem, not height.

③ $B = 144$, so $S_{total} = 384$.

④ $V = \frac{1}{3}(144)(8) = 384$

Height, not apothem.

ANSWER

$$m = 10, S_{lat} = 240, V = 384$$

EXAMPLE 2 Model answer, Q3: a cone of radius 6 and height 8.

① $l = \sqrt{36 + 64} = 10$

② $S_{curved} = \pi(6)(10) = 60\pi$

③ $V = \frac{1}{3}\pi(36)(8) = 96\pi$

④ $\theta = 360^\circ \times \frac{6}{10} = 216^\circ$

ANSWER

$$l = 10, 60\pi, 96\pi, 216^\circ$$

EXAMPLE 3 Model answer, Q5: a silo, cylinder $r = 3$, $h = 8$ m with a hemisphere, painted above ground.

$$\textcircled{1} \quad V = \pi(9)(8) + \frac{2}{3}\pi(27) = 72\pi + 18\pi = 90\pi$$

$$\approx 283 \text{ m}^3.$$

$$\textcircled{2} \quad \text{Curved cylinder } 2\pi(3)(8) = 48\pi.$$

$$\textcircled{3} \quad \text{Hemisphere } 2\pi(9) = 18\pi.$$

The two joined circles do not count.

$$\textcircled{4} \quad S = 66\pi \approx 207 \text{ m}^2.$$

Base excluded.

ANSWER

$$90\pi \approx 283 \text{ m}^3; 66\pi \approx 207 \text{ m}^2$$

EXAMPLE 4 Model answer, Q6: a closed tin of 500 cm^3 using least metal.

$$\textcircled{1} \quad S = 2\pi r^2 + \frac{1000}{r}$$

$$\textcircled{2} \quad \frac{dS}{dr} = 4\pi r - \frac{1000}{r^2} = 0$$

$$\textcircled{3} \quad r^3 = \frac{250}{\pi} \Rightarrow r \approx 4.30$$

$$\textcircled{4} \quad S'' > 0, \text{ so a minimum; } h \approx 8.60 = 2r.$$

ANSWER

$$r \approx 4.30 \text{ cm}, h \approx 8.60 \text{ cm}$$

03 Interactive model

Work Q5 twice — once counting all the faces, once dropping the joined ones — and let the class name the difference.

The year in twelve questions

Volume of a prism:

$$Ph$$

$$Bh$$

$$\frac{1}{3} Bh$$

$$2Bh$$

Volume of a cylinder:

$$2\pi rh$$

$$\pi r^2 h$$

$$\pi rh$$

$$\frac{1}{3} \pi r^2 h$$

Volume of a pyramid:

$$Bh$$

$$\frac{1}{3} Bh$$

$$\frac{1}{2} Bh$$

$$3Bh$$

Volume of a cone:

$$\pi r^2 h$$

$$\frac{1}{3} \pi r^2 h$$

$$\pi r l$$

$$\frac{1}{3} \pi r^2 l$$

Lateral area of a regular pyramid:

$$\frac{1}{2} Ph$$

$$\frac{1}{2} Pm$$

$$Pm$$

$$Bm$$

Curved area of a cone:

$$\pi r^2$$

$$\pi r l$$

$$\pi r h$$

$$2\pi r l$$

Volume of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$\frac{2}{3}\pi R^3$$

$$\pi R^3$$

Surface area of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$2\pi R^2$$

$$\pi R^2$$

Volume of a frustum of a cone:

$$\frac{\pi h}{3}(R^2 + r^2)$$

$$\frac{\pi h}{3}(R^2 + Rr + r^2)$$

$$\pi h(R^2 - r^2)$$

$$\frac{\pi h}{2}(R + r)$$

Joined faces of a compound solid:

count twice

count once

do not count

are halved

Scale factor k multiplies area by:

k

k^2

k^3

$2k$

Least-metal closed cylinder:

$$h = r$$

$$h = 2r$$

$$h = \frac{r}{2}$$

$$h = 3r$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Sphere	Sfera	Сфера
Compound solid	Murakkab jism	Составное тело
Optimisation	Optimallashtirish	Оптимизация

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q2 needs the apothem for:

the volume

the lateral area

the base

nothing

Q5 excludes:

the curved surfaces

the two joined circles and the base

nothing

the hemisphere

Q6 needs, beyond $f' = 0$:

nothing

a check that it is a minimum

a second constraint

a sketch

Lessons 61–68 are:

new material

revision and the entrance paper

a project

a test only

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume of a prism, $B = 15$, $h = 8$

ANSWER 120

2. Volume of a cylinder, $r = 2$, $h = 7$

ANSWER 28π

3. Volume of a pyramid, $B = 144$, $h = 8$

ANSWER 384

4. Volume of a cone, $r = 6$, $h = 8$

ANSWER 96π

5. Slant of that cone

ANSWER 10

6. Radius of a ball of volume 972π

ANSWER 9

7. Surface area of that ball

ANSWER 324π

Medium · the standard question

7 PROBLEMS

1. Hexagonal prism, edge 4, height 9: volume

ANSWER $216\sqrt{3} \approx 374$

2. Same prism: total area

ANSWER $48\sqrt{3} + 216 \approx 299$

3. Square pyramid, base 12, height 8: apothem and lateral area

ANSWER 10, 240

4. Same pyramid: volume

ANSWER 384

5. Cone $r = 6$, $h = 8$: curved area and sector angle

ANSWER 60π , 216°

6. Cap of height 3 on a ball of radius 9: curved area

ANSWER 54π

7. Silo, cylinder $r = 3$, $h = 8$ plus hemisphere: volume

ANSWER $90\pi \approx 283$

Hard · stretch and reason

7 PROBLEMS

1. Same silo: painted area, base excluded

ANSWER $66\pi \approx 207 \text{ m}^2$

2. Closed tin of 500 cm^3 : the radius of least metal

ANSWER $\approx 4.30 \text{ cm}$

3. Same tin: the least area

ANSWER $\approx 349 \text{ cm}^2$

4. Cap of height 3 on a ball of radius 9: its volume

ANSWER $\frac{\pi}{3}(9)(27 - 3) = 72\pi$

5. Two similar cones of volumes 27 and 64: the area ratio

ANSWER $9 : 16$

6. A cone and a hemisphere on the same base of radius r have equal volumes: find h

ANSWER $h = 2r$

7. A frustum of radii R and $\frac{R}{2}$: the fraction of the full cone

ANSWER $\frac{7}{8}$

07 Homework

Homework — 4 tasks

Bring the nine-formula page and the concept map to Lesson 61.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Write the nine formulas of the year on one page in your own hand, each with a small sketch.
3. A grain silo is a cylinder of radius 2.5 m and height 9 m with a hemispherical top. Find its volume and the area painted above ground.

4. A closed cylindrical tin holds 750 cm^3 . Find the radius that uses least metal, and verify that it is a minimum.

Revision — coordinates and vectors in space

[Cambridge revision]

Three numbers for a point, one product for every angle.

LESSON 61–62

2 × 40 MIN

GEOMETRY 11, CHAPTER I REVIEW

P2/P3 9.1–9.7

LESSON CLOCK

12'

20'

20'

16'

12'

Points and distances	12 min
The scalar product	20 min
Lines in space	20 min
Planes and normals	16 min
Homework	12 min

LEARNING OBJECTIVES

- Compute distances, midpoints and vectors in three dimensions.
- Use the scalar product to find an angle and to test perpendicularity.
- Write the vector equation of a line and decide whether two lines meet.
- Find the angle between a line and a plane using the normal.

TEXTBOOK

National	pp. 315–324
Cambridge	Pure Mathematics 2 & 3, pp. 190–223
Workbook	Exercise 9A–9G

01

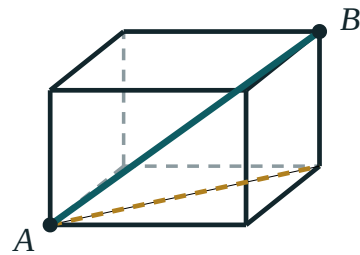
Explanation

Points and distances

Every plane formula gains a third term, and nothing else changes.

$$AB = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2} \quad M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2} \right)$$

Pythagoras
used twice



$$AB = \sqrt{(\Delta x^2 + \Delta y^2 + \Delta z^2)}$$

The space diagonal of a box — two applications of
Pythagoras, one after the other.

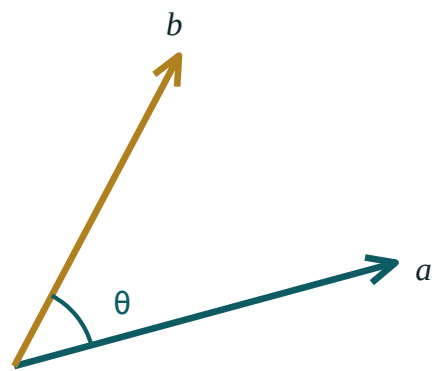
A	B	AB	M
(0,0,0)	(2,3,6)	7	(1, 1.5, 3)
(1,2,3)	(4,6,15)	13	(2.5, 4, 9)
(2,-1,4)	(6,2,8)	$\sqrt{41}$	(4, 0.5, 6)

A BOX OF EDGES a, b, c

Its space diagonal is $\sqrt{a^2 + b^2 + c^2}$ — the distance formula with the origin at one corner. Every three-dimensional distance question is this box, drawn somewhere.

The scalar product

$$u \cdot v = x_1x_2 + y_1y_2 + z_1z_2 = |u||v| \cos \theta$$



$a \cdot b = |a| \, |b| \, \cos \theta = a_1b_1 + a_2b_2 + a_3b_3$
perpendicular exactly when $a \cdot b = 0$

Two vectors and the angle between them — one product answers it.

$\cos \theta = \frac{u \cdot v}{|u||v|} \quad u \perp v \Leftrightarrow u \cdot v = 0$

<i>U</i>	<i>V</i>	<i>U</i> · <i>V</i>	θ
(1,2,2)	(2,-1,2)	4	$\approx 63.6^\circ$
(1,0,0)	(0,1,0)	0	90°
(3,4,0)	(6,8,0)	50	0°

WHY THIS REPLACES THE WHOLE OF GRADE 10’S ANGLE WORK

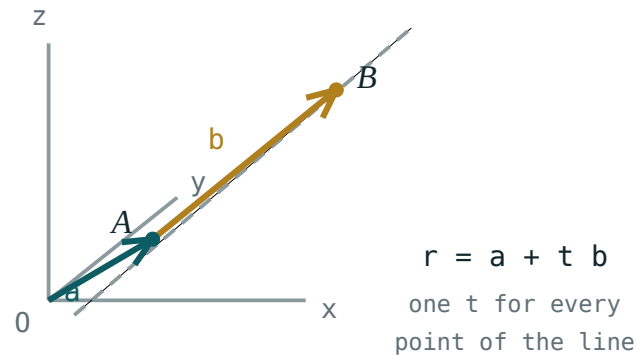
In Grade 10, an angle in space needed a projection, a right triangle and a careful drawing. Here it needs one dot product and one division. The picture is still worth drawing — but it is no longer the method.

U·*V* IS A NUMBER, NOT A VECTOR

Writing $u \cdot v = (2, 4, 4)$ confuses the scalar product with a scaling. The result of a dot product is always a single number, and its **sign** alone tells you whether the angle is acute (> 0), right ($= 0$) or obtuse (< 0).

Lines in space

$r = a + t \, d$ — through the point a , in the direction d



*A point and a direction — every other point of the line
is $a + t d$.*

Two lines in space may be parallel, may meet, or may be **skew**. The test is a system:

1. if d_1 is a multiple of d_2 , the lines are **parallel**;
2. otherwise set $a_1 + t d_1 = a_2 + s d_2$ — three equations in t and s ;
3. solve two of them; if the third is satisfied the lines **meet**, and if not they are **skew**.

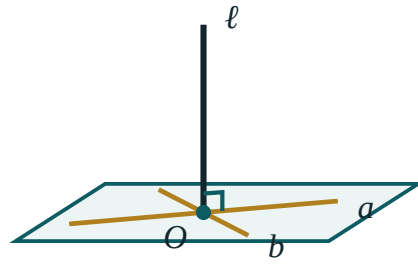
THE THIRD EQUATION IS THE WHOLE TEST

Two equations in two unknowns almost always have a solution. It is the **third** that decides, and forgetting to check it is why skew lines get reported as intersecting. Always substitute back.

The angle between two lines is the angle between their direction vectors — taken as the acute one, so use $|d_1 \cdot d_2|$.

Planes and normals

$$n \cdot r = n \cdot a, \text{ that is } ax + by + cz = d \text{ with normal } n = (a, b, c)$$



$\ell \perp a$ and $\ell \perp b$, with a and b intersecting at O
therefore ℓ is perpendicular to the whole plane

The normal — perpendicular to every line of the plane.

QUANTITY	FORMULA
angle between two planes	$\cos \theta = \frac{ \mathbf{n}_1 \cdot \mathbf{n}_2 }{ \mathbf{n}_1 \mathbf{n}_2 }$
angle between a line and a plane	$\sin \theta = \frac{ \mathbf{d} \cdot \mathbf{n} }{ \mathbf{d} \mathbf{n} }$
distance from a point to a plane	$\frac{ \mathbf{n} \cdot \mathbf{p} - d }{ \mathbf{n} }$

LINE AND PLANE USES **SINE**, NOT COSINE

The dot product gives the angle to the **normal**; the angle to the plane is its complement. So the formula carries *sin*, not *cos*. Using cosine gives the complement of the right answer — and $90^\circ - \theta$ is rarely one of the options.

Example. The angle between $\mathbf{d} = (1, 2, 2)$ and the plane $2x - y + 2z = 5$: $\mathbf{n} = (2, -1,$

$2)$, $\mathbf{d} \cdot \mathbf{n} = 2 - 2 + 4 = 4$, $|\mathbf{d}| = |\mathbf{n}| = 3$, so $\sin \theta = \frac{4}{9}$ and $\theta \approx 26.4^\circ$.

02 Worked examples

EXAMPLE 1 Find $|AB|$ and the midpoint for $A(1, 2, 3)$ and $B(4, 6, 15)$.

$$1 \quad AB = (3, 4, 12)$$

$$2 \quad |AB| = \sqrt{9 + 16 + 144} = 13$$

$$3 \quad M = (2.5, 4, 9)$$

ANSWER

13 and $(2.5, 4, 9)$

EXAMPLE 2 Find the angle between $u = (1, 2, 2)$ and $v = (2, -1, 2)$.

$$1 \quad u \cdot v = 2 - 2 + 4 = 4$$

$$2 \quad |u| = |v| = 3$$

$$3 \quad \cos \theta = \frac{4}{9}$$

$$4 \quad \theta \approx 63.6^\circ$$

ANSWER

$\approx 63.6^\circ$

EXAMPLE 3 Do the lines $r = (1, 0, 1) + t(2, 1, 1)$ and $r = (0, 1, 2) + s(1, -1, 0)$ meet?

- 1 $1 + 2t = s, t = 1 - s, 1 + t = 2$
- 2 From the third, $t = 1$; from the second, $s = 0$.
- 3 Check the first: $1 + 2 = 3 \neq 0$.
It fails.
- 4 The lines are skew.

ANSWER

Skew — the third equation is not satisfied

EXAMPLE 4 Find the angle between the line $r = a + t(1, 2, 2)$ and the plane $2x - y + 2z = 5$.

- 1 $n = (2, -1, 2)$
- 2 $d \cdot n = 2 - 2 + 4 = 4$
- 3 $\sin \theta = \frac{4}{3 \times 3} = \frac{4}{9}$
Sine, because n is the normal.
- 4 $\theta \approx 26.4^\circ$

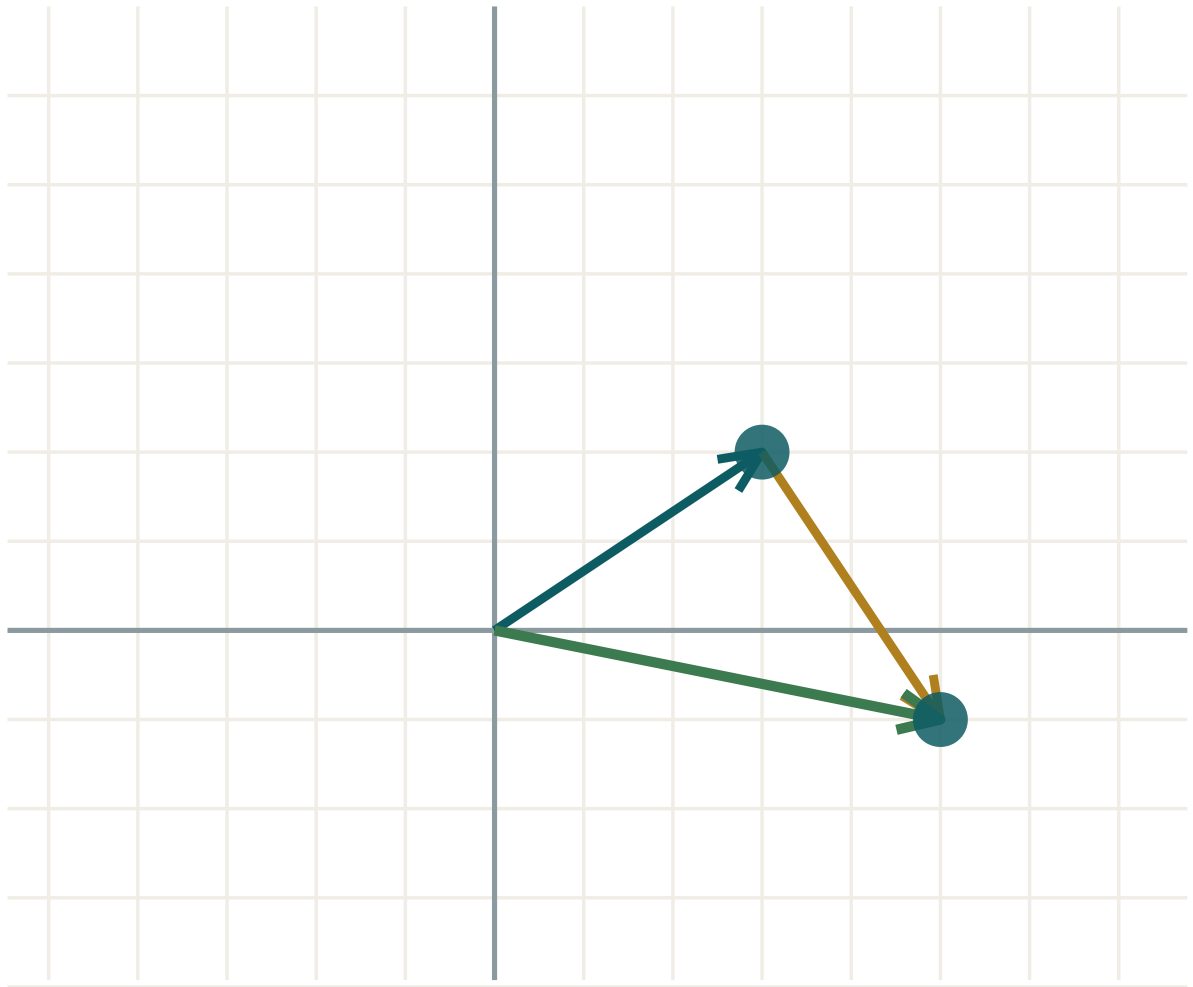
ANSWER

$\approx 26.4^\circ$

03 Interactive model

Use three metre rules taped at a corner of the room as the axes; every coordinate becomes a physical position.

Vectors and the angle between them



$$a \ (3; \ 2)$$

$$b \ (2; \ -3)$$

$$a + b \ (5; \ -1)$$

$$|a| \ 3.61$$

$$|b| \ 3.61$$

$$|a + b| \ 5.1$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinates in space	Fazoviy koordinatalar	Координаты в пространстве
Position vector	Radius-vektor	Радиус-вектор
Unit vector	Birlik vektor	Единичный вектор
Scalar (dot) product	Skalyar ko'paytma	Скалярное произведение
Perpendicular vectors	Perpendikulyar vektorlar	Перпендикулярные векторы
Vector equation of a line	To'g'ri chiziqning vektor tenglamasi	Векторное уравнение прямой
Direction vector	Yo'naltiruvchi vektor	Направляющий вектор
Normal vector	Normal vektor	Вектор нормали
Skew lines	Ayqash chiziqlar	Скрещивающиеся прямые
Parameter	Parametr	Параметр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

$|AB|$ in space is:

$$\sqrt{\Delta x^2 + \Delta y^2}$$

$$\sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}$$

$$\Delta x + \Delta y + \Delta z$$

$$\Delta x \Delta y \Delta z$$

$u \cdot v$ is:

a vector

a number

an angle

a length

$u \perp v$ when:

$$u \cdot v = 1$$

$$u \cdot v = 0$$

$$u = v$$

$$|u| = |v|$$

A negative dot product means the angle is:

acute

right

obtuse

zero

A line in space is written:

$$y = mx + c$$

$$r = a + t d$$

$$ax + by = c$$

$$n \cdot r = d$$

Two lines are skew when:

they are parallel

two of three equations hold but not the third

they meet

they coincide

The angle between a line and a plane uses:

$\cos$

$\sin$

$\tan$

$\cot$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $|AB|$ for $A(0,0,0)$, $B(2,3,6)$

ANSWER 7

2. Midpoint of $(1,2,3)$ and $(4,6,15)$

ANSWER $(2.5, 4, 9)$

3. $(1,2,2) \cdot (2,-1,2)$

ANSWER 4

4. $|(3,4,12)|$

ANSWER 13

5. Are $(1,0,0)$ and $(0,1,0)$ perpendicular?

ANSWER Yes — dot product 0

6. Space diagonal of a $3 \times 4 \times 12$ box

ANSWER 13

7. Direction vector of $r = (1,0,1) + t(2,1,1)$

ANSWER $(2, 1, 1)$

Medium · the standard question

7 PROBLEMS

1. Angle between $(1,2,2)$ and $(2,-1,2)$

ANSWER $\approx 63.6^\circ$

2. $|AB|$ for $A(2,-1,4)$, $B(6,2,8)$

ANSWER $\sqrt{41} \approx 6.40$

3. Unit vector along $(2,3,6)$

ANSWER $\frac{1}{7}(2,3,6)$

4. Do $r = (1,0,1) + t(2,1,1)$ and $r = (0,1,2) + s(1,-1,0)$ meet?

ANSWER No — skew

5. Angle between the line $d = (1,2,2)$ and the plane $2x - y + 2z = 5$

ANSWER $\approx 26.4^\circ$

6. Normal of the plane $3x + 4y - 12z = 7$

ANSWER $(3, 4, -12)$

7. Distance from $(1, 1, 1)$ to $2x - y + 2z = 5$

ANSWER $\frac{2}{3} \approx 0.667$

Hard · stretch and reason

7 PROBLEMS

1. Angle between the planes $x + y = 1$ and $y + z = 1$

ANSWER 60°

2. Shortest distance between the two skew lines above

ANSWER $\frac{3}{\sqrt{11}} \approx 0.905$

3. A cube of edge a with a vertex at the origin: the angle between two space diagonals

ANSWER $\approx 70.5^\circ$

4. Find k so that $(1, k, 2)$ is perpendicular to $(3, -1, 1)$

ANSWER $k = 5$

5. The plane through $(1,0,0)$, $(0,1,0)$, $(0,0,1)$

ANSWER $x + y + z = 1$

6. Angle between the line $r = t(1,1,1)$ and that plane

ANSWER 90°

7. Show that $(1,2,2)$, $(2,1,-2)$ and $(2,-2,1)$ are mutually perpendicular

ANSWER All three dot products vanish

07 Homework

Homework — 5 tasks

Sketch the box for every distance question; the third coordinate is easy to lose.

- Find $|AB|$ and the midpoint for $A(-1, 3, 2)$ and $B(5, -1, 10)$.
- Find the angle between $(2, 1, 2)$ and $(1, -2, 2)$, and say whether it is acute or obtuse.
- Decide whether the lines $r = (1,1,0) + t(1,2,1)$ and $r = (2,0,1) + s(2,1,-1)$ meet, are parallel or are skew.
- Find the angle between the line with direction $(2, 2, 1)$ and the plane $x + 2y + 2z = 9$.

5. Find k so that $(3, k, -2)$ is perpendicular to $(1, 4, 5)$, and check with the dot product.

Revision — parallelism and perpendicularity in space

The Grade 10 statements, revisited with a year of solids behind them.

LESSON 63–64

2 × 40 MIN

GEOMETRY 10–11 REVIEW

SPATIAL REVIEW

LESSON CLOCK

12'

20'

20'

16'

12'

The statements	12 min
Angles, both ways	20 min
Distances	20 min
Choosing a method	16 min
Homework	12 min

LEARNING OBJECTIVES

- State the positions of lines and planes, and the criteria for parallelism and perpendicularity.
- Use the three-perpendiculars theorem to reduce a space problem to a plane one.
- Compute angles between lines, between a line and a plane, and between planes.
- Choose between the synthetic and the vector method for each question.

TEXTBOOK

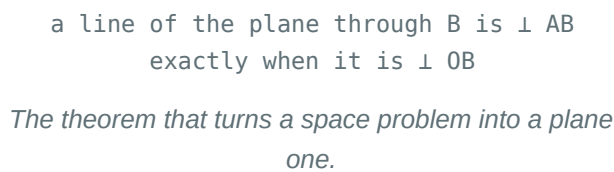
National	pp. 325–334
Cambridge	Pure Mathematics 2 & 3, pp. 281–284
Workbook	Revision set G3

01

Explanation

The statements

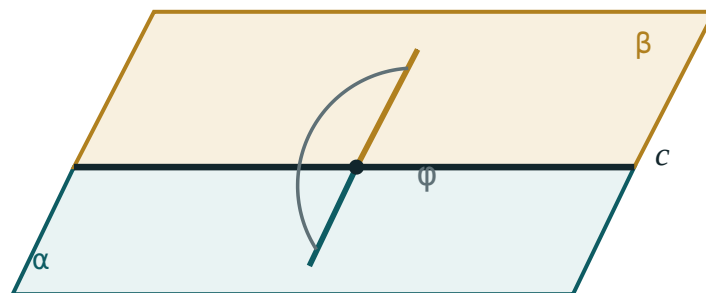
STATEMENT	CONDITION
a line is parallel to a plane	it is parallel to some line in the plane
two planes are parallel	two intersecting lines of one are parallel to two of the other
a line is perpendicular to a plane	it is perpendicular to two intersecting lines of it
two planes are perpendicular	one contains a line perpendicular to the other
three perpendiculars	a line in a plane $\perp$ an oblique $\Leftrightarrow \perp$ its projection



Two *parallel* lines of one plane parallel to two of another prove nothing — a line and a plane through it satisfy that with no parallelism at all. The same word carries the same weight in the perpendicularity criterion.

Angles, both ways

ANGLE	SYNTHETIC METHOD	VECTOR METHOD
two lines	translate one to meet the other	$\cos \theta = \frac{ \mathbf{d}_1 \cdot \mathbf{d}_2 }{ \mathbf{d}_1 \mathbf{d}_2 }$
line and plane	angle with the projection	$\sin \theta = \frac{ \mathbf{d} \cdot \mathbf{n} }{ \mathbf{d} \mathbf{n} }$
two planes	the linear angle at the edge	$\cos \theta = \frac{ \mathbf{n}_1 \cdot \mathbf{n}_2 }{ \mathbf{n}_1 \mathbf{n}_2 }$



both arms are perpendicular to the edge c
the angle φ between them is the dihedral angle

The linear angle — both rays perpendicular to the edge.

The same question, both ways. In a cube $ABCD A_1 B_1 C_1 D_1$ of edge 1, the angle between AC_1 and the base:

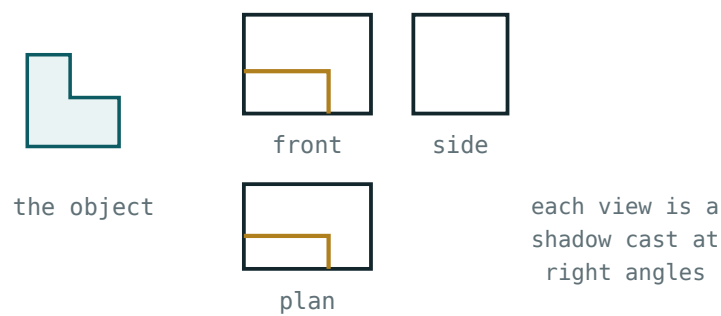
SYNTHETIC	VECTOR
projection $AC = \sqrt{2}$, height 1, $\tan \theta = \frac{1}{\sqrt{2}}$	$d = (1,1,1)$, $n = (0,0,1)$, $\sin \theta = \frac{1}{\sqrt{3}}$
$\theta \approx 35.3^\circ$	$\theta \approx 35.3^\circ$

BOTH GIVE THE SAME NUMBER, AND BOTH ARE WORTH FULL MARKS

The synthetic method needs a good drawing; the vector method needs good coordinates. Choose by which you can set up faster — but check the vector answer against a rough sketch, because a sign error in n is invisible in the algebra and obvious in the picture.

Distances

DISTANCE	HOW
point to plane	the perpendicular; or $\frac{ \mathbf{n} \cdot \mathbf{p} - d }{ \mathbf{n} }$
point to line	the perpendicular in the plane containing both
parallel planes	any point of one to the other
skew lines	the common perpendicular



three orthogonal projections determine the solid

Every distance in space is the length of one perpendicular.

Standard results in a cube of edge a .

DISTANCE	VALUE
between two opposite faces	a
between two skew edges	a
from a vertex to the far space diagonal	$\frac{a\sqrt{6}}{3}$
from the centre to a face	$\frac{a}{2}$
from a vertex to the plane A_1BD	$\frac{a\sqrt{3}}{3}$

THE VOLUME TRICK FOR A POINT-TO-PLANE DISTANCE

For a tetrahedron, $V = \frac{1}{3} Bh$ gives $h = \frac{3V}{B}$. Computing the volume two ways — once with an easy base, once with the awkward one — finds the distance without any construction at all. It is the fastest route in almost every cube question.

Choosing a method

THE QUESTION GIVES	PREFER	BECAUSE
a cube or a box	vectors	the coordinates write themselves
a proof, no numbers	synthetic	the criteria are the argument
a regular pyramid	synthetic	one right triangle answers it
a plane by an equation	vectors	the normal is already there
skew lines	vectors	the common perpendicular is hard to draw
a distance in a cube	the volume trick	no construction needed

ONE SENTENCE TO REMEMBER THE WHOLE REVISION BY

Every three-dimensional problem is solved by finding the one right triangle it contains — or, in vector form, the one dot product it depends on. Two methods, one idea: reduce the space question to something two-dimensional.

02 Worked examples

EXAMPLE 1 In a cube of edge 1, find the angle between AC_1 and the base, both ways.

- ① Synthetic: projection $AC = \sqrt{2}$, height 1.

$$\tan \theta = \frac{1}{\sqrt{2}}$$

- ② Vector: $d = (1,1,1)$, $n = (0,0,1)$.

③ $\sin \theta = \frac{1}{\sqrt{3}}$

- ④ $\theta \approx 35.3^\circ$
Both agree.

ANSWER

$\approx 35.3^\circ$

EXAMPLE 2 In a cube of edge a , find the distance from A to the plane A_1BD .

① Tetrahedron AA_1BD : $V = \frac{1}{6}a^3$.

Three mutually perpendicular edges.

② Face A_1BD is equilateral of side $a\sqrt{2}$.

③ $B = \frac{\sqrt{3}}{4}(2a^2) = \frac{a^2\sqrt{3}}{2}$

④ $h = \frac{3V}{B} = \frac{a}{\sqrt{3}} = \frac{a\sqrt{3}}{3}$

The volume trick.

ANSWER

$$\frac{a\sqrt{3}}{3} \approx 0.577a$$

EXAMPLE 3 Find the angle between the planes $x + y = 1$ and $y + z = 1$.

① $n_1 = (1, 1, 0), n_2 = (0, 1, 1)$

② $n_1 \cdot n_2 = 1$

③ $\cos \theta = \frac{1}{2}$

④ $\theta = 60^\circ$

ANSWER

$$60^\circ$$

EXAMPLE A point is 12 cm from a plane. Two obliques of 13 and 20 cm are drawn. Compare their projections and the angles they make with the plane.

① Projections 5 and 16.
 $\sqrt{169 - 144}$ and $\sqrt{400 - 144}$.

② $\sin \theta_1 = \frac{12}{13} \Rightarrow \theta_1 \approx 67.4^\circ$

③ $\sin \theta_2 = \frac{12}{20} \Rightarrow \theta_2 \approx 36.9^\circ$

④ The longer oblique makes the smaller angle.

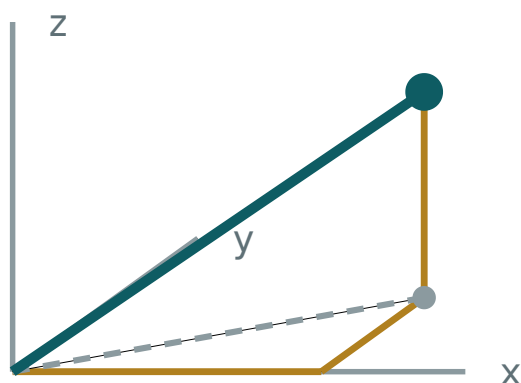
ANSWER

5 and 16 cm; $\approx 67.4^\circ$ and $\approx 36.9^\circ$

03 Interactive model

Set up the cube with coordinates on the board and solve one question twice; the class times both methods.

The cube, its diagonals and its planes



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Skew lines	Ayqash to'g'ri chiziqlar	Скрещивающиеся прямые
Parallel planes	Parallel tekisliklar	Параллельные плоскости
Perpendicular to a plane	Tekislikka perpendikulyar	Перпендикулярная плоскости
Three perpendiculars theorem	Uch perpendikulyar teoremasi	Теорема о трёх перпендикулярах
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Linear angle	Chiziqli burchak	Линейный угол
Orthogonal projection	Ortogonal proyeksiya	Ортогональная проекция
Synthetic method	Sintetik usul	Синтетический метод
Vector method	Vektor usuli	Векторный метод
Common perpendicular	Umumiy perpendikulyar	Общий перпендикуляр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

A line $\perp$ to a plane is $\perp$ to:

The angle to a plane uses the vector formula with:

The distance from a vertex of a cube of edge a to the plane A_1BD :

For a cube question, the faster method is usually:

For a proof with no numbers, prefer:

The longer oblique from a point makes:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Three positions of two lines in space

ANSWER Intersecting, parallel, skew

2. A line $\perp$ to two intersecting lines of a plane is:

ANSWER Perpendicular to the plane

3. Space diagonal of a cube of edge a

ANSWER $a\sqrt{3}$

4. Face diagonal of the same

ANSWER $a\sqrt{2}$

5. Projection of a 13 oblique from a point 12 away

ANSWER 5

6. Distance between two opposite faces of a cube of edge a

ANSWER a

7. Normal of $x + y = 1$

ANSWER $(1, 1, 0)$

Medium · the standard question

7 PROBLEMS

1. Angle between AC_1 and the base of a cube

ANSWER $\approx 35.3^\circ$

2. Angle between the planes $x + y = 1$ and $y + z = 1$

ANSWER 60°

3. Distance from a vertex of a cube of edge a to the plane A_1BD

ANSWER $\frac{a\sqrt{3}}{3}$

4. Angles of obliques 13 and 20 from a point 12 from a plane

ANSWER $\approx 67.4^\circ, \approx 36.9^\circ$

5. Angle between two space diagonals of a cube

ANSWER $\approx 70.5^\circ$

6. Distance between two skew edges of a cube of edge a

ANSWER a

7. Dihedral angle ABC_1 with the base of a cube

ANSWER 45°

Hard · stretch and reason

7 PROBLEMS

1. Distance from a vertex of a cube of edge a to the far space diagonal

ANSWER $\frac{a\sqrt{6}}{3}$

2. A regular tetrahedron of edge a : the angle between a face and the base

ANSWER $\approx 70.5^\circ$

3. Same tetrahedron: the distance from a vertex to the opposite face

ANSWER $\frac{a\sqrt{6}}{3}$

4. A regular square pyramid, base 12, height 8: the dihedral angle along a base edge

ANSWER $\approx 53.1^\circ$

5. Same pyramid: the angle of a lateral edge with the base

ANSWER $\approx 43.3^\circ$

6. Shortest distance between $r = t(1,0,0)$ and $r = (0,1,1) + s(0,1,0)$

ANSWER 1

7. Prove that the three face diagonals from one vertex of a cube are not coplanar

ANSWER Their scalar triple product is $-2a^3 \neq 0$

07 Homework

Homework — 5 tasks

Solve at least two questions by both methods and record which was faster.

1. State the five criteria of the first table, each with a sketch.
2. In a cube of edge 6 cm, find the angle between the space diagonal and the base by both methods.
3. Find the angle between the planes $2x - y + z = 3$ and $x + y + 2z = 1$.

4. A point is 9 cm from a plane; obliques of 15 and 41 cm are drawn. Find both projections and both angles.
5. Find the distance from a vertex of a cube of edge 4 cm to the plane through the three adjacent vertices, using the volume trick.

Revision — surface areas and volumes of all the solids

Nine formulas, one page, and the three questions that decide which of them a problem wants.

LESSON 65–66

2 × 40 MIN

GEOMETRY 11, FULL MENSURATION REVIEW

IGX 13.X FULL REVIEW

LESSON CLOCK

12'

16'

20'

20'

12'

The nine formulas	12 min
The three questions	16 min
Compound and hollow	20 min
Scale factors	20 min
A mixed set	12 min

LEARNING OBJECTIVES

- Recall and apply every surface-area and volume formula of the two-year course.
- Decide from the wording which measure a problem wants.
- Handle compound and hollow solids, and scale factors.
- Work a mixed set at examination pace.

TEXTBOOK

National	pp. 335–344
Cambridge	Core & Extended, pp. 397–404
Workbook	Revision set G4

01

Explanation

The nine formulas

SOLID	VOLUME	LATERAL / CURVED	TOTAL
prism	Bh	Ph	$2B + Ph$
cylinder	$\pi r^2 h$	$2\pi rh$	$2\pi r(r + h)$
pyramid (regular)	$\frac{1}{3} Bh$	$\frac{1}{2} Pm$	$B + \frac{1}{2} Pm$
cone	$\frac{1}{3} \pi r^2 h$	πrl	$\pi r(r + l)$
frustum of a cone	$\frac{\pi h}{3} (R^2 + Rr + r^2)$	$\pi l(R + r)$	+ two circles
sphere	$\frac{4}{3} \pi R^3$	$4\pi R^2$	$4\pi R^2$
hemisphere	$\frac{2}{3} \pi R^3$	$2\pi R^2$	$3\pi R^2$

THREE PATTERNS WORTH MORE THAN THE NINE LINES

- Every **pointed** solid is a third of the matching flat-topped one: $\frac{1}{3}$ for pyramid and cone.
- Every **lateral** area is “perimeter times a length, halved if pointed”: Ph and $\frac{1}{2} Pm$; $2\pi rh$ and πrl .
- Every **curved** area of a cone or pyramid uses the **slant**; every volume uses the **height**.

Learn the three patterns and the nine formulas rebuild themselves.

The three questions

BEFORE CHOOSING A FORMULA

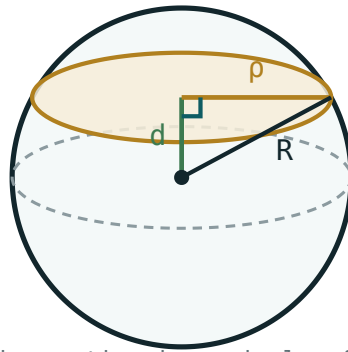
1. **Volume or area?** The words are “hold”, “fill”, “weigh” against “paint”, “cover”, “label”.
2. **Which faces?** Open or closed, solid or hollow, standing on the ground or not.
3. **Which length?** Height for volume, slant for curved area, radius not diameter.

WORDING	FORMULA
“a label around a tin”	$2\pi rh$
“a lid for a tin”	πr^2
“the canvas of a conical tent, no floor”	πrl
“the concrete in a solid pyramid”	$\frac{1}{3} Bh$
“the water in a hemispherical bowl”	$\frac{2}{3} \pi R^3$
“the glass in a hollow sphere”	$\frac{4}{3} \pi (R^3 - r^3)$

Compound and hollow

$$V_{\text{compound}} = V_1 + V_2 \quad S_{\text{compound}} = S_1 + S_2 - 2 \times (\text{shared face})$$

OBJECT	VOLUME	OUTSIDE AREA
silo: cylinder + hemisphere	$\pi r^2 h + \frac{2}{3} \pi r^3$	$2\pi rh + 2\pi r^2 (+ \text{base?})$
ice-cream: cone + hemisphere	$\frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$	$\pi rl + 2\pi r^2$
pipe: cylinder – cylinder	$\pi(R^2 - r^2)L$	$2\pi(R + r)L + 2 \text{ rings}$
hollow sphere	$\frac{4}{3} \pi (R^3 - r^3)$	$4\pi(R^2 + r^2)$



the section is a circle of radius ρ
 $\rho^2 = R^2 - d^2$

A plane cut through a solid of revolution — the shared face is always a circle.

THE TWO SHARED CIRCLES VANISH, NOT ONE

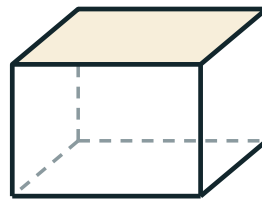
When a hemisphere sits on a cylinder, the cylinder's top circle and the hemisphere's flat face are both internal. Subtracting only one of them leaves πr^2 too much every time.

Scale factors, and a mixed set

$length \times k \quad area \times k^2 \quad volume \times k^3$



$k = 1$



$k = 2$

lengths $\times k$ · areas $\times k^2$ · volumes $\times k^3$

One scale factor, three different powers.

Read backwards this is just as useful: from a volume ratio take the cube root to get k , then square it for areas.

#	QUESTION	ANSWER
1	Volume of a cone, $r = 6$, $h = 8$	96π
2	Total area of that cone	96π
3	Volume of a sphere of radius 9	972π
4	Surface area of that sphere	324π
5	Volume of a square pyramid, base 10, height 12	400
6	Its lateral area	260
7	Frustum, $R = 5$, $r = 3$, $h = 6$	98π
8	Two similar solids, areas 9 : 25: volume ratio	27 : 125

QUESTION 2 IS THE TRAP

A cone with $r = 6$ and $h = 8$ has $l = 10$, so its total area is $\pi r(r + l) = 96\pi$ — numerically equal to its volume, purely by coincidence of these numbers. Recognising that the two answers **should** differ in units is the check that catches a formula used in the wrong place.

02 Worked examples

EXAMPLE 1 A cone has $r = 6$ and $h = 8$. Find its volume and total surface area.

$$(1) \quad l = \sqrt{36 + 64} = 10$$

$$(2) \quad V = \frac{1}{3}\pi(36)(8) = 96\pi$$

Height.

$$(3) \quad S = \pi(6)(6 + 10) = 96\pi$$

Slant.

$$(4) \quad \text{Equal numbers, different units — cm}^3 \text{ against cm}^2.$$

ANSWER

$96\pi \text{ cm}^3$ and $96\pi \text{ cm}^2$

EXAMPLE 2 A hollow glass sphere has outer radius 10 cm and inner 9 cm. Find the volume of glass.

$$(1) \quad V = \frac{4}{3}\pi(1000 - 729)$$

$$(2) \quad = \frac{4}{3}\pi(271)$$

$$(3) \quad \approx 1135 \text{ cm}^3.$$

ANSWER

$$\frac{1084\pi}{3} \approx 1135 \text{ cm}^3$$

EXAMPLE 3 Two similar cones have surface areas in the ratio 9 : 25. Find the ratio of their volumes.

$$① \quad k^2 = \frac{25}{9} \Rightarrow k = \frac{5}{3}$$

Square root first.

$$② \quad k^3 = \frac{125}{27}$$

$$③ \quad 27 : 125$$

ANSWER

27 : 125

EXAMPLE 4 An ice-cream is a cone of radius 3 and height 4 with a hemisphere on top. Find its volume and outside surface area.

$$① \quad V = \frac{1}{3} \pi (9)(4) + \frac{2}{3} \pi (27) = 12\pi + 18\pi = 30\pi$$

$$② \quad l = 5, \text{ so cone curved } \pi(3)(5) = 15\pi.$$

No base circle — it is joined.

$$③ \quad \text{Hemisphere curved } 2\pi(9) = 18\pi.$$

Its flat face is joined too.

$$④ \quad S = 33\pi \approx 104 \text{ cm}^2.$$

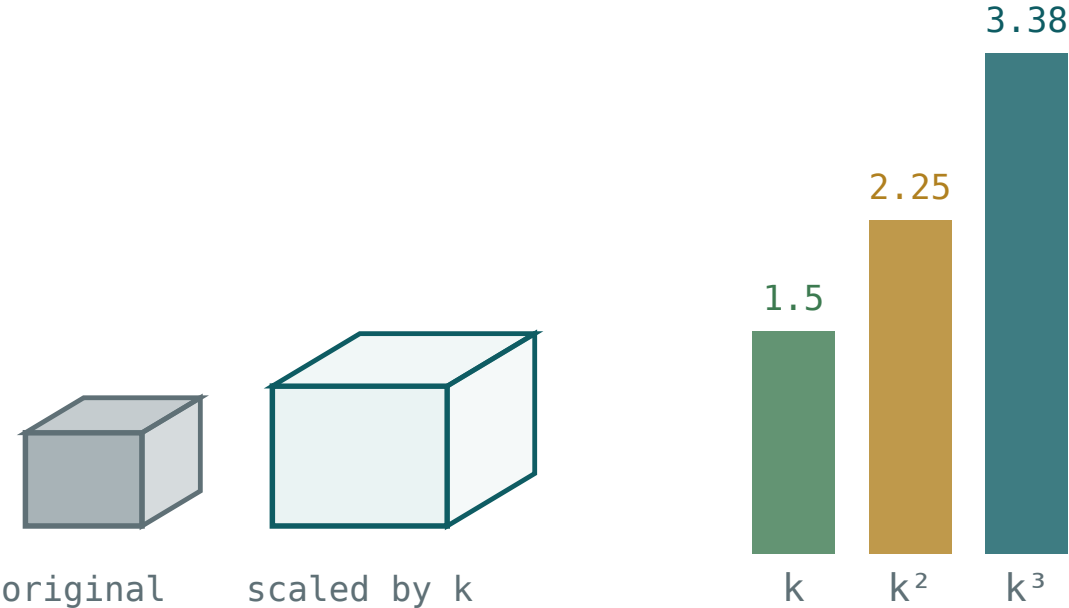
ANSWER

$30\pi \approx 94.2 \text{ cm}^3$; $33\pi \approx 104 \text{ cm}^2$

03 Interactive model

Put the nine formulas on the board, then rub out three at random and have the class rebuild them from the three patterns.

One factor, three powers



k 1.5	length × 1.5	area ×k² 2.25	volume ×k³ 3.38
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Cylinder	Silindr	Цилиндр
Pyramid	Piramida	Пирамида
Cone	Konus	Конус
Frustum	Kesik jism	Усечённое тело
Sphere	Sfera	Сфера
Lateral surface	Yon sirt	Боковая поверхность
Total surface	To'la sirt	Полная поверхность
Scale factor	O'xshashlik koeffitsiyenti	Коэффициент подобия
Compound solid	Murakkab jism	Составное тело

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Volume of a pyramid:

Bh

$\frac{1}{3} Bh$

$\frac{1}{2} Bh$

$3Bh$

Curved area of a cone:

πr^2

$\pi r l$

$\pi r h$

$2\pi r l$

Volume of a cone:

$\pi r^2 h$

$\frac{1}{3} \pi r^2 h$

$\frac{1}{3} \pi r^2 l$

$\pi r l$

Total area of a hemisphere (solid):

$$2\pi R^2$$

$$3\pi R^2$$

$$4\pi R^2$$

$$\pi R^2$$

Volume of a frustum of a cone:

$$\frac{\pi h}{3}(R^2 + r^2)$$

$$\frac{\pi h}{3}(R^2 + Rr + r^2)$$

$$\pi h(R^2 - r^2)$$

$$\frac{\pi h}{2}(R + r)$$

Areas in the ratio 9 : 25 give volumes in:

$$9 : 25$$

$$27 : 125$$

$$3 : 5$$

$$81 : 625$$

Two joined faces of a compound solid:

both count

one counts

neither counts

they are halved

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume of a prism, $B = 20$, $h = 7$

ANSWER 140

2. Volume of a cylinder, $r = 3$, $h = 5$

ANSWER 45π

3. Volume of a cone, $r = 6$, $h = 8$

ANSWER 96π

4. Slant of that cone

ANSWER 10

5. Volume of a sphere, $R = 9$

ANSWER 972π

6. Surface area of that sphere

ANSWER 324π

7. Volume of a pyramid, $B = 100$, $h = 12$

ANSWER 400

Medium · the standard question

7 PROBLEMS

1. Total area of a cone, $r = 6$, $h = 8$

ANSWER 96π

2. Lateral area of a square pyramid, base 10, height 12

ANSWER 260

3. Frustum volume, $R = 5$, $r = 3$, $h = 6$

ANSWER 98π

4. Total area of a cylinder, $r = 3$, $h = 5$

ANSWER 48π

5. Total area of a solid hemisphere, $R = 4$

ANSWER 48π

6. Areas 9 : 25: the volume ratio

ANSWER 27 : 125

7. Volumes 8 : 27: the area ratio

ANSWER 4 : 9

Hard · stretch and reason

7 PROBLEMS

1. Hollow glass sphere, $R = 10$, $r = 9$: volume of glass

ANSWER $\approx 1135 \text{ cm}^3$

2. Ice-cream, cone $r = 3$, $h = 4$ plus hemisphere: volume and outside area

ANSWER $30\pi; 33\pi$

3. Silo, cylinder $r = 4$, $h = 12$ plus hemisphere: volume

ANSWER $\frac{704\pi}{3} \approx 737$

4. Pipe $R = 6$, $r = 4$, $L = 150$: volume of metal

ANSWER $3000\pi \approx 9425$

5. A cone of radius 9 cut halfway up: volume of the frustum as a fraction

ANSWER $\frac{7}{8}$

6. A sphere inscribed in a cylinder: the ratio of total areas

ANSWER $2 : 3$

7. A cube and a sphere of equal volume: the ratio of surface areas (cube : sphere)

ANSWER $6 : \sqrt[3]{36\pi} \approx 1 : 0.806$

07 Homework

Homework — 5 tasks

Answer the three questions in writing before each calculation; that is the whole revision.

1. Write the nine formulas from memory, then check them and mark any you got wrong.
2. A cone has radius 5 cm and height 12 cm. Find its volume, curved area and total area.
3. A hollow metal sphere has outer radius 8 cm and inner 7 cm. Find the volume of metal and its mass at 7.8 g/cm^3 .

4. Two similar solids have volumes 64 and 125 cm^3 . Find the ratio of their heights and of their surface areas.
5. A grain hopper is a cylinder of radius 2 m and height 5 m with a cone of height 3 m below it. Find its capacity in litres.

Geometry in the entrance paper, and the annual review

What the examination actually asks, and what two years of geometry come to.

LESSON 67–68

2 × 40 MIN

GEOMETRY 11, YAKUNIY TAKRORLASH

EXAMINATION TECHNIQUE

LESSON CLOCK

12'

24'

16'

16'

12'

The seven types	12 min
A timed mixed set	24 min
Diagnosis by type	16 min
The two years in one page	16 min
The plan	12 min

LEARNING OBJECTIVES

- Recognise the geometry question types of the entrance paper.
- Choose the fastest of the synthetic, coordinate and vector methods.
- Work a mixed timed set and diagnose the errors by type.
- Leave the course with a usable summary and a plan.

TEXTBOOK

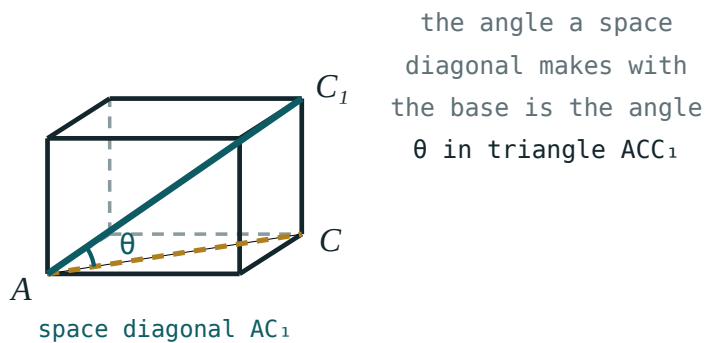
National	pp. 345–352
Cambridge	Core & Extended, pp. 405–408
Workbook	Revision set G5

01

Explanation

The seven types

TYPE	TYPICAL QUESTION	FASTEST METHOD
plane figure	a triangle's area or an angle	synthetic, with a sketch
circle	a tangent, a chord, an inscribed angle	coordinates if an equation is given
coordinate	a line, a circle, an intersection	algebra straight through
solid: measurement	a volume or an area	the nine formulas
solid: angle	a line and a plane, or a dihedral	vectors in a box, synthetic elsewhere
solid: section	name and measure a cut	synthetic, three rules
similarity	a ratio of lengths, areas or volumes	k, k^2, k^3



One box, and five of the seven types measured inside it.

THE FIRST LINE OF EVERY ANSWER

Name the solid or the figure, and write down what you know: B , h , r , the coordinates, the vectors. The question rarely names them for you, and the mark scheme rarely gives anything before that line.

A timed mixed set

Twelve questions, twenty minutes. Mark anything that takes over ninety seconds and come back to it.

#	QUESTION	ANSWER
1	Area of a triangle with sides 13, 14, 15	84
2	Tangent to $x^2 + y^2 = 25$ at (3, 4)	$3x + 4y = 25$
3	Distance from (1, 8) to $y = 2x$	$\frac{6}{\sqrt{5}} \approx 2.68$
4	Volume of a cone, $r = 5$, $h = 12$	100π
5	Curved area of that cone	65π
6	Space diagonal of a $3 \times 4 \times 12$ box	13
7	Angle of that diagonal with the base	$\approx 67.4^\circ$
8	Section of a cube through six edge-midpoints	regular hexagon
9	Volume of a sphere of surface area 144π	288π
10	Two similar solids, volumes 27 : 64: area ratio	9 : 16
11	Angle between (1,2,2) and (2,2,1)	$\approx 27.3^\circ$
12	Least-metal closed cylinder: $\frac{h}{r}$	2

Diagnosis by type

Do not count the score. Sort the errors.

ERROR TYPE	LOOKS LIKE	WHAT TO PRACTISE
reading	curved when total was asked	the three questions, written out
formula	a missing $\frac{1}{3}$ or $\frac{4}{3}$	the nine-formula page, daily
method	the long route to a short answer	the method table above
arithmetic	a right method, a wrong number	slow down on the last line
units	cm ³ where litres were wanted	convert lengths first

THE NUMBER THAT PREDICTS THE EXAM SCORE

Not the practice score — the number of errors of the **same type** repeated across two papers. One repeated type is worth four or five marks in the real examination, and each one takes about twenty minutes of targeted practice to remove.

The two years in one page, and the plan

GRADE 10	GRADE 11
axioms; three points determine a plane	coordinates; a point is three numbers
parallelism: parallel to a line in it	vectors; parallel means a scalar multiple
perpendicularity: two intersecting lines	the scalar product; $u \cdot v = 0$
angles by projection	angles by dot product
sections of polyhedra	prisms, pyramids, cones, spheres
the plane coordinate method	mensuration in context, and optimisation

ONE SENTENCE FOR THE WHOLE OF SCHOOL GEOMETRY

Every problem is solved by finding the one right triangle it contains — or, with vectors, the one dot product it depends on. Two years, one idea: reduce what is in front of you to something you can already measure.

The plan. Between now and the examination: the nine formulas rewritten from memory once a week; one timed mixed set a week, sorted by error type; and the list of repeated types kept on a single sheet. Not more mathematics — the same mathematics, faster and with fewer repeated slips.

That is the end of the geometry course. The formulas will fade; the habit of drawing the figure before writing anything will not, and it is the more useful of the two.

02 Worked examples

EXAMPLE 1 Model answer, №1: the area of a triangle with sides 13, 14, 15.

① $s = 21$

Heron.

② $S = \sqrt{21 \times 8 \times 7 \times 6}$

③ $= \sqrt{7056} = 84$

A standard triangle worth remembering.

ANSWER

84

EXAMPLE 2 Model answer, №7: the angle of the space diagonal of a $3 \times 4 \times 12$ box with the base.

① Base diagonal $\sqrt{9 + 16} = 5$.

② Space diagonal $\sqrt{25 + 144} = 13$.

③ $\tan \theta = \frac{12}{5}$

④ $\theta \approx 67.4^\circ$

ANSWER

$\approx 67.4^\circ$

EXAMPLE 3 Model answer, №11: the angle between $(1,2,2)$ and $(2,2,1)$.

$$1 \quad u \cdot v = 2 + 4 + 2 = 8$$

$$2 \quad |u| = |v| = 3$$

$$3 \quad \cos \theta = \frac{8}{9}$$

$$4 \quad \theta \approx 27.3^\circ$$

ANSWER

$$\approx 27.3^\circ$$

EXAMPLE 4 Model answer, №9: the volume of a sphere of surface area 144π .

$$1 \quad 4\pi R^2 = 144\pi \Rightarrow R = 6$$

$$2 \quad V = \frac{4}{3}\pi(216)$$

$$3 \quad = 288\pi \approx 905$$

Not 144π — area and volume are different powers.

ANSWER

$$288\pi$$

03 Interactive model

Run the twelve questions strictly to time, then spend the rest of the lesson only on the error-type table.

The two years in twelve questions

A plane is determined by:

two points

three non-collinear points

one line

four points

Two lines that never meet are:

parallel

parallel or skew

skew

intersecting

A line $\perp$ to a plane is $\perp$ to:

one line

two intersecting lines, then all

two parallel lines

the normal

The angle to a plane is measured to:

the normal

the projection

any line

the edge

Volume of a prism:

Ph

Bh

$\frac{1}{3} Bh$

$2Bh$

Volume of a cone:

$\pi r^2 h$

$\frac{1}{3} \pi r^2 h$

$\pi r l$

$\frac{1}{3} \pi r^2 l$

Curved area of a cone:

$$\pi r^2$$

$$\pi r l$$

$$\pi r h$$

$$2\pi r l$$

Surface area of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$2\pi R^2$$

$$\pi R^2$$

Volumes 27 : 64 give areas in:

$$27 : 64$$

$$9 : 16$$

$$3 : 4$$

$$81 : 256$$

A section of a cube has at most:

$u \perp v$ when:

 $u \cdot v = 1$

 $u \cdot v = 0$

 $u = v$

 $|u| = |v|$

Least-metal closed cylinder:

 $h = r$

 $h = 2r$

 $h = \frac{r}{2}$

 $h = 3r$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Entrance examination	Kirish imtihoni	Вступительный экзамен
Question type	Savol turi	Тип задачи
Synthetic method	Sintetik usul	Синтетический метод
Coordinate method	Koordinatalar usuli	Координатный метод
Vector method	Vektor usuli	Векторный метод
Time management	Vaqtni taqsimlash	Распределение времени
Error type	Xato turi	Тип ошибки
Summary sheet	Xulosa varaqasi	Итоговый лист

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first line of an entrance answer should be:

the formula

the named figure and what is known

the answer

a guess

For an angle inside a box, prefer:

synthetic

vectors

estimation

trial

After a practice paper, count:

the score

the repeated error types

the time

the blanks

The one idea of two years of geometry:

memorise formulas

reduce to a right triangle or a dot product

draw neatly

work fast

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Area of a triangle with sides $13, 14, 15$

ANSWER 84

2. Tangent to $x^2 + y^2 = 25$ at $(3, 4)$

ANSWER $3x + 4y = 25$

3. Volume of a cone, $r = 5, h = 12$

ANSWER 100π

4. Curved area of that cone

ANSWER 65π

5. Space diagonal of a $3 \times 4 \times 12$ box

ANSWER 13

6. Radius of a sphere of area 144π

ANSWER 6

7. Volumes $27 : 64$: area ratio

ANSWER $9 : 16$

Medium · the standard question

7 PROBLEMS

1. Angle of the $3 \times 4 \times 12$ space diagonal with the base

ANSWER $\approx 67.4^\circ$

2. Distance from $(1, 8)$ to $y = 2x$

ANSWER $\frac{6}{\sqrt{5}} \approx 2.68$

3. Section of a cube through six edge-midpoints

ANSWER A regular hexagon

4. Volume of a sphere of area 144π

ANSWER 288π

5. Angle between $(1,2,2)$ and $(2,2,1)$

ANSWER $\approx 27.3^\circ$

6. Least-metal closed cylinder: $\frac{h}{r}$

ANSWER 2

7. Total area of a cone, $r = 5$, $h = 12$

ANSWER 90π

Hard · stretch and reason

7 PROBLEMS

1. A cube of edge a : the distance from a vertex to the plane of the three adjacent vertices

ANSWER $\frac{a\sqrt{3}}{3}$

2. A regular tetrahedron of edge a : its volume

ANSWER $\frac{a^3\sqrt{2}}{12}$

3. A sphere inscribed in a cone of radius 6 and height 8

ANSWER $r = 3$

4. A cone of radius 9 cut halfway up: the frustum as a fraction of the cone

ANSWER $\frac{7}{8}$

5. The regular hexagonal section of a cube of edge a : its area

ANSWER $\frac{3a^2\sqrt{3}}{4}$

6. Two spheres of radii 3 and 5 touch externally: the distance between the centres

ANSWER 8

7. A closed cylinder of volume V with least area: express that area in terms of V

ANSWER $3\sqrt[3]{2\pi V^2}$

07 Homework

Homework — 4 tasks

This is the last geometry lesson of the school course. The summary page is the thing to keep.

1. Work the twelve-question set again under timed conditions and sort every error by type.
2. Write the two years in one page: six lines for Grade 10, six for Grade 11, with a sketch for each.
3. Write the nine formulas from memory, check them, and note which you got wrong.

4. Write your revision plan for the weeks before the examination in three sentences, and date it.